TAYLOR EXPANSION WITH MEMORY COMMUTES WITH OHANA TREES IN THE LAMBDA-I CALCULUS

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Allowed standard prerequisites (editorial).

- Standard lambda-calculus head reduction, Church-Rosser and λI normalization properties. Author §2 and reference Barendregt; complete source bibliography retained. Resource/Ohana-specific proofs are included rather than treated as background.

1. Preliminaries

We recall some basic notions and results concerning the λ-calculus, and its fragment without weakening known as the λI-calculus. For more information, we refer to Barendregt's book [4].

1.1. **The λ-calculus, in a nutshell.** Let us fix an infinite set $\mathcal{V}$ of variables. The set Λ of λ-terms is defined by induction

$$\Lambda \quad \ni \quad M, N ::= x \mid \lambda x.M \mid MN \quad (\text{for } x \in \mathcal{V})$$

We use parentheses when needed. We assume that application associates on the left, and has higher precedence than abstraction. E.g., $\lambda x.\lambda y.\lambda z.xyz$ stands for $\lambda x.(\lambda y.(\lambda z.(xy)z))$. We write $\lambda x_1 \dots x_n.M$ or even $\lambda \vec{x}.M$ for $\lambda x_1 \dots . \lambda x_n.M$, and $N^n(M)$ for $N(N(\cdots N(M)\cdots))$, where N occurs n times. In particular, when $n = 0$, we get $\lambda \vec{x}.M = N^n(M) = M$.

The set $\mathrm{fv}(M)$ of free variables of M is defined as usual, and we say that M is closed whenever $\mathrm{fv}(M) = \varnothing$. Hereafter, λ-terms are considered modulo α-conversion (see [4, §2.1]).

Example 1. The λ-terms below are used throughout the paper to construct examples:

$$\mathsf{I} \coloneqq \lambda x.x, \quad \mathsf{K} \coloneqq \lambda xy.x, \quad \mathsf{F} \coloneqq \lambda xy.y, \quad \mathsf{B} \coloneqq \lambda fgx.f(g(x)), \quad \mathsf{D} \coloneqq \lambda x.xx, \quad \Omega \coloneqq \mathsf{DD}.$$

The λ-term I is the identity, K, F are the projections, B is the composition, D is the diagonal and Ω a looping term. The pairing of M, N is encoded as $[M, N] \coloneqq \lambda x.xMN$, for x fresh.

The β-reduction $\to_\beta$ is generated by the rule $(\lambda x.M)N \to M[N/x]$, where $(\lambda x.M)N$ is called a β-redex and $M[N/x]$ stands for the capture-free substitution of N for all free occurrences of x in M. We let $\twoheadrightarrow_\beta$ (resp. $=_\beta$) be the reflexive-transitive (and symmetric) closure of $\to_\beta$. We say that M is in β-normal form (β-nf) if M does not contain any β-redex, and that

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M has a β-nf if $M \twoheadrightarrow_\beta N$, for some N in β-nf. Since $\to_\beta$ is confluent, such N must be unique.

Remark 2. Any λ-term M can be written in one of the following forms: either $M = \lambda\vec{x}.yM_1 \cdots M_k$, in which case M is called a head normal form (hnf) and y its head variable; or $M = \lambda\vec{x}.(\lambda y.P)QM_1 \cdots M_k$, where $(\lambda y.P)Q$ is referred to as its head redex.

The head reduction $\to_h$ is the restriction of $\to_\beta$ obtained by contracting the head redex.

Definition 3. A λ-term Y is a fixed-point combinator (fpc) if it satisfies $Yx =_\beta x(Yx)$.

Notice that we do not require that fixed-point combinators are closed λ-terms. We recall Curry's combinator Y and define, for all $l \in \mathcal{V}$, Polonsky's Θ_l and Klop's $\boldsymbol{\mathfrak{L}}_l$ [35]:

(1)

$$\mathsf{D}_f \coloneqq \lambda x.f(xx), \quad \mathsf{Y} \coloneqq \lambda f.\mathsf{D}_f\mathsf{D}_f, \quad \mathsf{V} \coloneqq \lambda vlf.f(vvlf), \quad \Theta_l \coloneqq \mathsf{V}\mathsf{V}l, \quad \boldsymbol{\mathfrak{L}}_l \coloneqq \lambda e.\mathsf{B}\mathsf{Y}\mathsf{B}el.$$

It is easy to check that Y and Θ_l are fpcs. Regarding the Bible $\boldsymbol{\mathfrak{L}}_l$, which owes its name to its distinctive spelling, we get $\boldsymbol{\mathfrak{L}}_l x =_\beta \mathsf{B}\mathsf{Y}\mathsf{B}xl =_\beta \mathsf{Y}(\mathsf{B}x)l =_\beta \mathsf{B}x(\mathsf{Y}(\mathsf{B}x))l =_\beta x(\mathsf{Y}(\mathsf{B}x)l) =_\beta x(\boldsymbol{\mathfrak{L}}_l x)$. Note that the variable l remains in a passive position along the reduction of $\boldsymbol{\mathfrak{L}}_l$ (and of Θ_l), therefore $\boldsymbol{\mathfrak{L}}_N \coloneqq \boldsymbol{\mathfrak{L}}_l[N/l]$ (resp. $\Theta_N \coloneqq \Theta_l[N/l]$) remains a fpc, for all $N \in \Lambda$.

Although fpcs Y are not β-normalizable, they differ from Ω since they produce increasing stable amounts of information along reduction: $Y \twoheadrightarrow_\beta \lambda f.fY_1 \twoheadrightarrow_\beta \lambda f.f^n(Y_n) \twoheadrightarrow_\beta \cdots$. Barendregt introduced a notion of 'evaluation tree' to capture this infinitary behaviour [3].

Definition 4. The Böhm tree $\mathrm{BT}(M)$ of a λ-term M is coinductively defined as follows.

(1) If M has a hnf, that is $M \twoheadrightarrow_h \lambda x_1 \dots x_n.yM_1 \cdots M_k$, then

$$\mathrm{BT}(M) \coloneqq \lambda x_1 \dots x_n.\, y \quad \text{with children } \mathrm{BT}(M_1) \cdots \mathrm{BT}(M_k)$$

(2) $\mathrm{BT}(M) \coloneqq \bot$, otherwise. The constant $\bot$ represents the absolute lack of information.

Example 5. Using the terms from Example 1 and (1), let us construct some Böhm trees.

(i) If M has a β-normal form N then, up to isomorphism, $\mathrm{BT}(M) = N$. E.g., $\mathrm{BT}(\mathsf{DI}) = \mathsf{I}$.

(ii) Since Ω and YK have no hnf, we get $\mathrm{BT}(\Omega) = \mathrm{BT}(\mathsf{YK}) = \bot$.

(iii) If Y is an fpc, then $\mathrm{BT}(Y) = \lambda f.f(f(f(\cdots)))$. Thus, $\mathrm{BT}(\mathsf{Y}) = \mathrm{BT}(\boldsymbol{\mathfrak{L}}_l) = \mathrm{BT}(\Theta_l)$.

(iv) Using the fpc Y, one can define λ-terms $\mathsf{E}_x, \mathsf{O}_x$ satisfying $\mathsf{E}_x =_\beta [\Omega, [\Omega x, \mathsf{E}_x]]$ and $\mathsf{O}_x =_\beta [\Omega x, [\Omega, \mathsf{O}_x]]$. These two λ-terms both construct a 'stream' $[M_1, [M_2, [M_3, [\cdots]]]]$, but the former places Ωx at even positions and Ω at odd ones, while the latter does the converse. This difference is however forgotten in their Böhm trees:

$$\mathrm{BT}(\mathsf{E}_x) = \mathrm{BT}(\mathsf{O}_x) = [\bot, [\bot, [\bot, [\bot, [\cdots]]]]] \text{ (inline depiction of a tree)}$$

(v) Check that, given $\mathsf{R} = \lambda ylx.x(y(xl))$, we have $\mathrm{BT}(\mathsf{YR}l) = \lambda x_0.x_0(\lambda x_1.x_1(\lambda x_2.x_2(\cdots)))$.

The Böhm tree equality induces an equivalence on λ-terms: $M =_{\mathcal{B}} N$ iff $\mathrm{BT}(M) = \mathrm{BT}(N)$. As shown in [31, 48], $\mathcal{B}$ is a λ-theory because it contains $=_\beta$ and is compatible with abstraction and application: $M =_{\mathcal{B}} M'$ entails $\lambda x.M =_{\mathcal{B}} \lambda x.M'$, $MN =_{\mathcal{B}} M'N$ and $NM =_{\mathcal{B}} NM'$.

1.2. **The $\lambda\mathsf{I}$-calculus.** The set Λ_{I} of $\lambda\mathsf{I}$-terms is the subset of Λ consisting of those λ-terms in which every abstraction binds at least one occurrence of the abstracted variable. Note that $\to_\beta$ induces a reduction on Λ_{I} since $M \in \Lambda_{\mathsf{I}}$ and $M \to_\beta N$ entail $N \in \Lambda_{\mathsf{I}}$. In other words, the property of being a $\lambda\mathsf{I}$-term is preserved by $\to_\beta$. Similarly, the $\lambda\mathsf{I}$-calculus inherits from λ-calculus all the notions previously defined like head reduction, Böhm trees, etc. For convenience, we consider an alternative definition of Λ_{I} simultaneously defining $\lambda\mathsf{I}$-terms and their sets of free variables.

Definition 6.

(i) For all $X \subseteq \mathcal{V}$, $\Lambda_{\mathsf{I}}(X)$ is defined as the smallest subset of Λ such that:

$$\frac{}{x \in \Lambda_{\mathsf{I}}(\{x\})} \qquad \frac{M \in \Lambda_{\mathsf{I}}(X) \quad x \in X}{\lambda x.M \in \Lambda_{\mathsf{I}}(X - \{x\})} \qquad \frac{M \in \Lambda_{\mathsf{I}}(X) \quad N \in \Lambda_{\mathsf{I}}(Y)}{MN \in \Lambda_{\mathsf{I}}(X \cup Y)}$$

(ii) Finally, the set of all $\lambda\mathsf{I}$-terms is given by the disjoint union $\Lambda_{\mathsf{I}} = \coprod_{X \subseteq \mathcal{V}} \Lambda_{\mathsf{I}}(X)$.

Remark 7.

(i) Note that $M \in \Lambda_{\mathsf{I}}(X)$ means that M is a $\lambda\mathsf{I}$-term such that $\mathrm{fv}(M) = X$. As a consequence, if $M \in \Lambda_{\mathsf{I}}(X)$ then the set X must be finite.
(ii) Each set $\Lambda_{\mathsf{I}}(X)$ is closed under β-conversion.
(iii) Our examples $\mathsf{I}, \mathsf{B}, \mathsf{D}, \Omega, \mathsf{Y}, \mathsf{H}_l, \Theta_l, \mathsf{E}_x, \mathsf{O}_x, \mathsf{YR}l$ are $\lambda\mathsf{I}$-terms, while $\mathsf{K}, \mathsf{F}, \mathsf{YK} \notin \Lambda_{\mathsf{I}}$.

Definition 8. A $\lambda\mathsf{I}$-theory $\mathcal{T}$ is given by an equivalence $=_{\mathcal{T}} \subseteq \Lambda_{\mathsf{I}}^2$ containing β-conversion and compatible with application and abstraction. In the context of $\lambda\mathsf{I}$-calculus, the compatibility with abstraction becomes: $M =_{\mathcal{T}} M'$ and $x \in \mathrm{fv}(M) \cap \mathrm{fv}(M')$ imply $\lambda x.M =_{\mathcal{T}} \lambda x.M'$.

The only subtlety in the definition above stands in the side condition on the compatibility with abstraction. However, this small difference has significant consequences: e.g., the theory axiomatized by equating all terms without a β-nf is consistent as a $\lambda\mathsf{I}$-theory, but inconsistent as a λ-theory. The following theorem collects the main properties of the $\lambda\mathsf{I}$-calculus.

Theorem 9.

(i) Every computable numerical function is $\lambda\mathsf{I}$-definable [4, Thm. 9.2.16].
(ii) Strong and weak β-normalization coincide for the $\lambda\mathsf{I}$-calculus [4, Thm. 11.3.4].
(iii) Every λ-theory is also a $\lambda\mathsf{I}$-theory, while the converse does not hold.

By (iii), Böhm trees induce a $\lambda\mathsf{I}$-theory, but we argue that this theory does not respect the spirit of Λ_{I}. E.g., the variable l is never erased along the reduction of, say, H_l, but it disappears in its Böhm tree. Using the terminology of [4], one says that l is pushed to infinity in $\mathrm{BT}(\mathsf{H}_l)$. This shows that the Böhm tree of the $\lambda\mathsf{I}$-term $\lambda l.\mathsf{H}_l \in \Lambda_{\mathsf{I}}$ is not a $\lambda\mathsf{I}$-tree: the outer λ-abstraction 'λl' does not bind any free occurrence of l in $\mathrm{BT}(\mathsf{H}_l) = \lambda f.f(f(f(\cdots)))$.

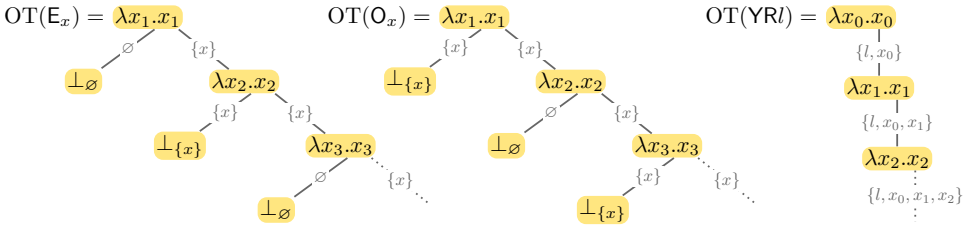


Figure 1. Examples of Ohana trees of λI-terms.

2. Ohana trees

We have seen that Böhm trees are not well-suited for representing λI-terms faithfully, because variables that are never erased along the reduction are forgotten (i.e., pushed into infinity). We now introduce a notion of evaluation tree that records all variables that are pushed along each path, ensuring that none are forgotten, whether they remain in passive position or not.

2.1. **The coinductive definition.** For every finite set $X \subseteq \mathcal{V}$, written $X \subseteq_{\mathrm{f}} \mathcal{V}$, we introduce a constant $\bot_X$ representing any looping $M \in \Lambda_{\mathsf{I}}(X)$. Intuitively, the Ohana tree of a λI-term M is obtained from $\mathrm{BT}(M)$ by annotating each subtree (also $\bot$) with the free variables of the λ-term that generated it.

Definition 10. The Ohana tree of a term $M \in \Lambda_{\mathsf{I}}(X)$ is coinductively defined as follows:

(i) If $M \twoheadrightarrow_h \lambda x_1 \dots x_n.yM_1 \cdots M_k$ with $y \in X \cup \{\vec{x}\}$, $M_i \in \Lambda_{\mathsf{I}}(X_i)$ (for $1 \le i \le k$), then

$$\mathrm{OT}(M) \coloneqq \lambda x_1 \dots x_n.y \quad \text{with children } \mathrm{OT}(M_1), \dots, \mathrm{OT}(M_k) \text{ on edges } X_1, \dots, X_k$$

(ii) $\mathrm{OT}(M) \coloneqq \bot_X$, otherwise. Recall that $X = \mathrm{fv}(M)$, since $M \in \Lambda_{\mathsf{I}}(X)$.

We sometimes use an inline presentation of Ohana trees, by introducing application symbols annotated with the sets X_i, as in $\lambda x_1 \dots x_n.y \cdot^{X_1} T_1 \cdots^{X_k} T_k$.

Example 11.

(i) $\mathrm{OT}(\mathsf{I}) = \lambda x.x$, $\mathrm{OT}(\mathsf{D}) = \lambda x.x \cdot^{\{x\}} x$, $\mathrm{OT}(\mathsf{B}) = \lambda fgx.f \cdot^{\{g,x\}} (g \cdot^{\{x\}} x)$.

(ii) $\mathrm{OT}(\Omega) = \mathrm{OT}(\lambda y.\Omega y) = \bot_{\varnothing}$, $\mathrm{OT}(\Omega x) = \bot_{\{x\}}$. More generally: $\mathrm{OT}(\Omega\vec{x}) = \bot_{\{x_1,\dots,x_n\}}$.

(iii) $\mathrm{OT}(\mathsf{Y}) = \lambda f.f \cdot^{\{f\}} (f \cdot^{\{f\}} (f \cdot^{\{f\}} (\cdots)))$. In fact, it is easy to check that for all closed fpcs $Y \in \Lambda_{\mathsf{I}}$, we have $\mathrm{OT}(Y) = \mathrm{OT}(\mathsf{Y}) \cong \mathrm{BT}(\mathsf{Y})$.

(iv) In $\mathrm{OT}(\mathsf{Y}_l)$ the variable l which is pushed into infinity becomes visible: $\mathrm{OT}(\mathsf{Y}_l) = \mathrm{OT}(\Theta_l) = \lambda f.f \cdot^{\{f,l\}} (f \cdot^{\{f,l\}} (f \cdot^{\{f,l\}} (\cdots)))$. Thus $\mathrm{OT}(\mathsf{Y}_l) \neq \mathrm{OT}(\Theta_x)$ when $x \neq l$.

Other examples of Ohana trees are depicted in Figure 1. Note that the Ohana trees of E_x and O_x are now distinguished. The interest of $\mathsf{YR}l$ is that it pushes into infinity an increasing amount of variables:

$$\mathsf{YR}l =_\beta \lambda x_0.x_0((\mathsf{YR})(x_0 l)) =_\beta \lambda x_0.x_0(\lambda x_1.x_1((\mathsf{YR})(x_1(x_0 l)))) =_\beta \cdots$$

So—at the limit—infinitely many variables are pushed into infinity (but only l is free).

Remark 12. If $M \in \Lambda_{\mathsf{I}}$ has a β-nf N, then there is an isomorphism $\mathrm{OT}(M) \cong N$. In fact, since N is finite, it is possible to reconstruct all the labels in $\mathrm{OT}(M)$.

We are going to show that Ohana trees are invariant under β-reduction (Prop. 19 plus Thm. 20), and that the equality induced on $\lambda\mathsf{I}$-terms is a $\lambda\mathsf{I}$-theory (Corollary 45).

2.2. **The associated approximation theory.** We introduce a theory of program approximation that captures Ohana trees, by adapting the well-established approach originally developed for Böhm trees [31, 48] (see also [4, Ch. 14]). We start by defining the approximants with memory, corresponding to finite Ohana trees.

Definition 13.

(i) The set of approximants with memory is $\mathcal{A}_{\mathsf{m}} \coloneqq \coprod_{X \subseteq_f \mathcal{V}} \mathcal{A}_{\mathsf{m}}(X)$, where $\mathcal{A}_{\mathsf{m}}(X)$ is defined by (for $A_i \in \mathcal{A}_{\mathsf{m}}(X_i)$, $X_i \subseteq X$, $y \in X \cup \{\vec{x}\}$ and $\vec{x} \in \bigcup_{i=1}^{k} X_i$):

$$\mathcal{A}_{\mathsf{m}}(X) \quad \ni \quad A ::= \bot_X \mid \lambda x_1 \dots x_n.y A_1 \cdots A_k$$

(ii) We define $\sqsubseteq \subseteq \mathcal{A}_{\mathsf{m}}^2$ as the least partial order closed under the following rules:

$$\frac{A \in \mathcal{A}_{\mathsf{m}}(X)}{\bot_X \sqsubseteq A} \qquad \frac{A_i \sqsubseteq A_i' \quad A_i, A_i' \in \mathcal{A}_{\mathsf{m}}(X_i) \quad \text{for all } 1 \le i \le k \quad \vec{x} \in \{y\} \cup \left(\bigcup_{i=1}^{k} X_i\right)}{\lambda x_1 \dots x_n.y\, A_1 \cdots A_k \sqsubseteq \lambda x_1 \dots x_n.y\, A_1' \cdots A_k'}$$

Each $(\mathcal{A}_{\mathsf{m}}(X), \sqsubseteq)$ is a pointed poset (partially ordered set) with bottom element $\bot_X$, while the poset $(\mathcal{A}_{\mathsf{m}}, \sqsubseteq)$ is not pointed because it has countably many minimal elements. Note that we do not annotate the application symbols in $A \in \mathcal{A}_{\mathsf{m}}$ with the sets of variables X_i, since the labels in its $\bot$-nodes carry enough information to reconstruct the finite Ohana tree associated with A.

Lemma 14. There is a bijection between $\mathcal{A}_{\mathsf{m}}$ and the set of finite Ohana trees of $\lambda\mathsf{I}$-terms.

We now give the recipe to compute the set of approximants with memory of a $\lambda\mathsf{I}$-term.

Definition 15.

(i) The direct approximant of $M \in \Lambda_{\mathsf{I}}(X)$ is given by:

$$\omega_{\mathsf{m}}(M) \coloneqq \begin{cases} \lambda x_1 \dots x_n.y\, \omega_{\mathsf{m}}(M_1) \cdots \omega_{\mathsf{m}}(M_k), & \text{if } M = \lambda x_1 \dots x_n.y M_1 \cdots M_k, \\ \bot_X, & \text{otherwise.} \end{cases}$$

(ii) The set $\mathrm{App}_{\mathsf{m}}(M)$ of approximants with memory of $M \in \Lambda_{\mathsf{I}}$ is defined by:

$$\mathrm{App}_{\mathsf{m}}(M) \coloneqq \{A \in \mathcal{A}_{\mathsf{m}} \mid \exists N \in \Lambda_{\mathsf{I}}\,.\, M \twoheadrightarrow_\beta N \text{ and } A \sqsubseteq \omega_{\mathsf{m}}(N)\}$$

Remark 16. For all $M \in \Lambda_{\mathsf{I}}(X)$, we have $\omega_{\mathsf{m}}(M) \in \mathcal{A}_{\mathsf{m}}(X)$ and $\mathrm{App}_{\mathsf{m}}(M) \subseteq \mathcal{A}_{\mathsf{m}}(X)$.

Example 17.

(i) $\mathrm{App}_{\mathsf{m}}(\mathsf{D}) = \{\bot_\varnothing, \lambda x.x\bot_{\{x\}}, \mathsf{D}\}$, $\mathrm{App}_{\mathsf{m}}(\Omega) = \{\bot_\varnothing\}$.

(ii) $\mathrm{App}_{\mathsf{m}}(\mathsf{Y}) = \{\bot_\varnothing\} \cup \{\lambda f.f^n(\bot_{\{f\}}) \mid n \in \mathbb{N}\}$.

(iii) $\mathrm{App}_{\mathsf{m}}(\boxplus_l) = \{\bot_{\{l\}}\} \cup \{\lambda f.f^n(\bot_{\{f,l\}}) \mid n \in \mathbb{N}\}$.
(iv) $\mathrm{App}_{\mathsf{m}}(\mathsf{Y}\mathsf{R}l) = \{\bot_{\{l\}}, \lambda x_0.x_0(\bot_{\{x_0,l\}}), \lambda x_0.x_0(\lambda x_1.x_1(\bot_{\{x_0,x_1,l\}})), \ldots\}$.

Lemma 18.

(i) For every $M, N \in \Lambda_{\mathsf{I}}$, $M \to_\beta N$ entails $\omega_{\mathsf{m}}(M) \sqsubseteq \omega_{\mathsf{m}}(N)$.
(ii) For all $M \in \Lambda_{\mathsf{I}}(X)$, $\mathrm{App}_{\mathsf{m}}(M)$ is an ideal of the poset $(\mathcal{A}_{\mathsf{m}}(X), \sqsubseteq)$. More precisely:
  (a) non-emptiness: $\bot_X \in \mathrm{App}_{\mathsf{m}}(M)$;
  (b) downward closure: $A \in \mathrm{App}_{\mathsf{m}}(M)$ and $A' \sqsubseteq A$ imply $A' \in \mathrm{App}_{\mathsf{m}}(M)$;
  (c) directedness: $\forall A_1, A_2 \in \mathrm{App}_{\mathsf{m}}(M), \exists A_3 \in \mathrm{App}_{\mathsf{m}}(M)$ such that $A_1 \sqsubseteq A_3 \sqsupseteq A_2$.

Proof. (i) By an easy induction on the structure of $M \in \Lambda_{\mathsf{I}}(X)$, using Remarks 2 and 7.
(ii) (a) and (b) are trivial. For (c), define $\sqcup$ by setting: $\bot_X \sqcup A = A \sqcup \bot_X = A$, $x \sqcup x = x$, if A_1, A_1' and A_2, A_2' are compatible, then $(\lambda x.A_1) \sqcup (\lambda x.A_1') = \lambda x.(A_1 \sqcup A_1')$ and $A_1A_2 \sqcup A_1'A_2' = (A_1 \sqcup A_1')(A_2 \sqcup A_2')$. Check that $A_3 = A_1 \sqcup A_2$ is their supremum. □

The next result shows that β-convertible terms share the same the set of approximants.

Proposition 19. Let $M, N \in \Lambda_{\mathsf{I}}(X)$. If $M \to_\beta N$, then $\mathrm{App}_{\mathsf{m}}(M) = \mathrm{App}_{\mathsf{m}}(N)$.

Proof. By confluence of $\to_\beta$, using Lemma 18(i). □

Theorem 20. There is a bijection $\mathcal{A}(\cdot)$ between the set of Ohana trees and the set of approximants of $\lambda\mathsf{I}$-terms satisfying $\mathcal{A}(\mathrm{OT}(M)) = \mathrm{App}_{\mathsf{m}}(M)$, for all $M \in \Lambda_{\mathsf{I}}$.

Proof. From $\mathrm{OT}(M)$ to $\mathrm{App}_{\mathsf{m}}(M)$. Let $\mathrm{OT}(M)^\star$ be the set of all finite subtrees of $\mathrm{OT}(M)$, where each truncated subtree T is replaced by $\bot_{\mathrm{fv}(T)}$. By Lemma 14, $\mathrm{OT}(M)^\star \cong \mathrm{App}_{\mathsf{m}}(M)$.

From $\mathrm{App}_{\mathsf{m}}(M)$ to $\mathrm{OT}(M)$. For every finite path σ in $\mathrm{OT}(M)$, there is an $A \in \mathrm{App}_{\mathsf{m}}(M)$ representing $\mathrm{OT}(M)$ along σ. Conclude since $\mathrm{OT}(M)$ is the supremum of all such σ. □

3. A resource calculus with memory

In this section we introduce the $\lambda\mathsf{I}$-resource calculus refining the (finite) resource calculus [19], best known as the target language of Ehrhard and Regnier's Taylor expansion [21]. So, the resource calculus is not meant to be a stand-alone language, but rather another theory of approximations for the λ-calculus. Before going further, we recall its main properties.

We consider here the promotion-free fragment of the resource calculus introduced in [39]. Its syntax is similar to the λ-calculus, except for the applications that are of shape $s\bar{t}$, where $\bar{t} = [t_1, \ldots, t_n]$ is a multiset of resources called bag. The resources populating $\bar{t}$ are linear as they cannot be erased or copied by s, they must be used exactly once along the reduction. When contracting a term of the form $s = (\lambda x.s')[t_1, \ldots, t_n]$ there are two possibilities.

(1) If the number of occurrences of x in s' is exactly n, then each occurrence is substituted by a different t_i. Since the elements in the bag are unordered, there is no canonical bijection between the resources and the occurrences of x. The solution consists

in collecting all possibilities in a formal sum of terms, the sum representing an inner-choice operator.

(2) If there is a mismatch between the number of occurrences and the amount of resources, then s reduces to the empty-sum, $\mathbf{0}$. From a programming-language perspective, this can be thought of as a program terminating abruptly after throwing an uncaught exception.

The first-class citizens of the resource calculus are therefore finite sums of resource terms, that are needed to ensure the (strong) confluence of reductions. Another important property is strong normalization, that follows from the fact that no resource can be duplicated.

3.1. **$\lambda\mathsf{I}$-resource expressions and $\lambda\mathsf{I}$-resource sums.** Our version of the resource calculus is extended with labels representing the memory of free variables that were present in the $\lambda\mathsf{I}$-term they approximate. Just like in Theorem 13(i) we endowed the constant $\bot$ with a finite set X of variables, here we annotate:

- the empty bag 1 of resource terms, since an empty bag of approximants of M should remember the free variables of M;
- the empty sum $\mathbf{0}$ of resource terms. Indeed, if a resource term vanishes during reduction because of the mismatch described above, its free variables should be remembered.

Definition 21.

(i) For all $X \subseteq_{\mathrm{f}} \mathcal{V}$, the sets $\Delta_{\mathsf{I}}(X)$ and $!\Delta_{\mathsf{I}}(X)$ are defined by induction:

$$\frac{}{x \in \Delta_{\mathsf{I}}(\{x\})} \qquad \frac{s \in \Delta_{\mathsf{I}}(X) \quad x \in X}{\lambda x.s \in \Delta_{\mathsf{I}}(X - \{x\})} \qquad \frac{s \in \Delta_{\mathsf{I}}(X) \quad \bar{t} \in !\Delta_{\mathsf{I}}(Y)}{s\bar{t} \in \Delta_{\mathsf{I}}(X \cup Y)}$$

$$\frac{}{1_X \in !\Delta_{\mathsf{I}}(X)} \qquad \frac{t_0 \in \Delta_{\mathsf{I}}(X) \quad \cdots \quad t_n \in \Delta_{\mathsf{I}}(X)}{[t_0, \dots, t_n] \in !\Delta_{\mathsf{I}}(X)}$$

(ii) The set Δ_{I} of $\lambda\mathsf{I}$-resource terms and the set $!\Delta_{\mathsf{I}}$ of bags are given by:

$$\Delta_{\mathsf{I}} \coloneqq \coprod_{X \subseteq_{\mathrm{f}} \mathcal{V}} \Delta_{\mathsf{I}}(X) \quad \text{and} \quad !\Delta_{\mathsf{I}} \coloneqq \coprod_{X \subseteq_{\mathrm{f}} \mathcal{V}} !\Delta_{\mathsf{I}}(X).$$

As a matter of notation, we let $(!)\Delta_{\mathsf{I}}$ denote either Δ_{I} or $!\Delta_{\mathsf{I}}$, indistinctly but coherently. We call resource expressions generic elements $s, t \in (!)\Delta_{\mathsf{I}}$. We denote the union of two bags $\bar{t}, \bar{u} \in !\Delta_{\mathsf{I}}(X)$ multiplicatively by $\bar{t} \cdot \bar{u}$, whose neutral element is the empty bag, 1_X.

Remark 22.

(i) In every $\lambda\mathsf{I}$-resource term of the form $\lambda x.s$, the variable x must occur freely in s: it may appear in the undecorated underlying term, or in the "memory" X decorating 1_X. Therefore, it makes sense to define $\mathrm{fv}(s) \coloneqq X$ whenever $s \in (!)\Delta_{\mathsf{I}}(X)$.

(ii) Each set $!\Delta_{\mathsf{I}}(X)$ is isomorphic to the monoid of multisets of elements of $\Delta_{\mathsf{I}}(X)$. Notice that $!\Delta_{\mathsf{I}}$ is not the set of all bags of elements of Δ_{I}, just its subset of bags whose elements have the same free variables (and so inductively in the subterms).

Example 23.

(i) The identity I belongs to Δ_{I}, whereas the projections do not: $\mathsf{K}, \mathsf{F} \notin \Delta_{\mathsf{I}}$.

(ii) Note that $\lambda xy.x1_{\varnothing} \notin \Delta_{\mathsf{I}}$ since $y \notin \mathrm{fv}(x1_{\varnothing})$, but $\lambda xy.x1_{\{y\}} \in \Delta_{\mathsf{I}}$ because $y \in \{y\}$.
(iii) The terms $\mathsf{D}_0 = \lambda x.x1_{\{x\}}$ and $\mathsf{D}_{n+1} = \lambda x.x[x,\dots,x]$, where the bag contains $n+1$ occurrences of x, are $\lambda\mathsf{I}$-resource terms. By Remark 22(ii), we obtain:
(iv) $[x,y] \notin !\Delta_{\mathsf{I}}$, while $[x, x[x], x[\lambda y.y, \lambda y.y[y]], (\lambda y.y)[x], \lambda y.y1_{\{x\}}] \in !\Delta_{\mathsf{I}}(\{x\}) \subseteq !\Delta_{\mathsf{I}}$.

We consider resource expressions up to α-equivalence, under the proviso that abstractions bind linear occurrences of variables as well as occurrences in the memory of empty bags. For instance, $\lambda xyz.x1_{\{x\}}1_{\{x,y,z\}}$ and $\lambda x'y'z.x'1_{\{x'\}}1_{\{x',y',z\}}$ are considered α-equivalent.

Definition 24. For all $X \subseteq_{\mathrm{f}} \mathcal{V}$, the set $\mathbb{N}[(!)\Delta_{\mathsf{I}}(X)]$ of sums of $\lambda\mathsf{I}$-resource terms ("resource sums", for short) is defined as the $\mathbb{N}$-semimodule of finitely supported formal sums of expressions in $(!)\Delta_{\mathsf{I}}(X)$, with coefficients in $\mathbb{N}$. Explicitly, it can be presented as:

$$\mathbb{N}[(!)\Delta_{\mathsf{I}}(X)] \quad \ni \quad \mathbf{s},\mathbf{t} ::= \mathbf{0}_X \mid r \mid r + \mathbf{s} \quad (\text{for } r \in (!)\Delta_{\mathsf{I}}(X))$$

quotiented by associativity and commutativity of $+$, as well as neutrality of $\mathbf{0}_X$.

Note that $(!)\Delta_{\mathsf{I}}(X) \subseteq \mathbb{N}[(!)\Delta_{\mathsf{I}}(X)]$, that is, resource expressions are assimilated to the corresponding one-element sum. The constructors of the calculus are extended to resource sums by (bi)linearity, i.e. for $s \in \Delta_{\mathsf{I}}(X)$, $\mathbf{s} \in \mathbb{N}[\Delta_{\mathsf{I}}(X)]$, $\bar{t} \in !\Delta_{\mathsf{I}}(Y)$, $\bar{\mathbf{t}} \in \mathbb{N}[!\Delta_{\mathsf{I}}(Y)]$, $u \in \Delta_{\mathsf{I}}(Y)$ and $\mathbf{u} \in \mathbb{N}[\Delta_{\mathsf{I}}(Y)]$, we have:

$$(s+\mathbf{s})\bar{t} \coloneqq s\bar{t} + \mathbf{s}\bar{t}, \quad \mathbf{0}_X\bar{t} \coloneqq \mathbf{0}_{X\cup Y}, \quad \lambda x.(s+\mathbf{s}) \coloneqq \lambda x.s + \lambda x.\mathbf{s}, \quad \lambda x.\mathbf{0}_X \coloneqq \mathbf{0}_{X-\{x\}},$$
$$s(\bar{t}+\bar{\mathbf{t}}) \coloneqq s\bar{t} + s\bar{\mathbf{t}}, \quad s\mathbf{0}_Y \coloneqq \mathbf{0}_{X\cup Y}, \quad [u+\mathbf{u}]\cdot\bar{t} \coloneqq [u]\cdot\bar{t} + [\mathbf{u}]\cdot\bar{t}, \quad [\mathbf{0}_Y]\cdot\bar{t} \coloneqq \mathbf{0}_Y.$$

Therefore, if $\mathbf{0}_X$ occurs in s not as a summand but as a proper subterm, then $s = \mathbf{0}_{\mathrm{fv}(s)}$.

3.2. **Memory substitution and resource substitution.** While the usual finite resource calculus is completely linear, the variables we store in the memory of 1_X and $\mathbf{0}_X$ are not. The memory X remembers the variables present in X not their amounts, this is the reason why it is modelled as a set, not as a multiset. This consideration leads us to define two kinds of substitutions.

The (non-linear) memory substitution of a set $Y \subseteq_{\mathrm{f}} \mathcal{V}$ for a variable x in a $\lambda\mathsf{I}$-resource term s does not interact with the linear occurrences of x in s (i.e., they remain unchanged), it just replaces the "memory" of x in the empty bags with the content of Y.

Definition 25.

(i) For all $X, Y \subseteq_{\mathrm{f}} \mathcal{V}$ and $x \in \mathcal{V}$, define

$$X\{Y/x\} \coloneqq \begin{cases} X - \{x\} \cup Y, & \text{if } x \in X, \\ X, & \text{otherwise.} \end{cases}$$

(ii) Given $s \in (!)\Delta_{\mathsf{I}}(X)$, the memory substitution of x by $Y \subseteq_{\mathrm{f}} \mathcal{V}$ in s is the resource term $s\{Y/x\}$ defined as follows (for $x \neq y$ and, in the abstraction case, $y \notin Y$):

$$x\{Y/x\} \coloneqq x, \qquad 1_X\{Y/x\} \coloneqq 1_{X\{Y/x\}}, \qquad (s\bar{t})\{Y/x\} \coloneqq (s\{Y/x\})(\bar{t}\{Y/x\}),$$
$$y\{Y/x\} \coloneqq y, \quad (\lambda y.s)\{Y/x\} \coloneqq \lambda y.s\{Y/x\}, \quad ([s]\cdot\bar{t})\{Y/x\} \coloneqq [s\{Y/x\}]\cdot(\bar{t}\{Y/x\}).$$

We now define the resource substitution of a bag $\bar{u}$ for x in s, whose effect is twofold. 1) It non-deterministically replaces each linear occurrence of x with a resource from the bag (as usual). 2) It applies the memory substitution of x by $\mathrm{fv}(\bar{u})$ to the resulting sum of terms.

Definition 26. Given $s \in (!)\Delta_{\mathsf{I}}(X)$, $x \in \mathcal{V}$ and $\bar{u} \in !\Delta_{\mathsf{I}}(Y)$, the resource substitution of x by $\bar{u}$ in s is the resource sum $s\langle \bar{u}/x\rangle \in \mathbb{N}[(!)\Delta_{\mathsf{I}}(X\{Y/x\})]$ defined as follows:

$$x\langle \bar{u}/x\rangle \coloneqq \begin{cases} u, & \text{if } \bar{u}=[u],\\ \mathbf{0}_Y, & \text{otherwise,}\end{cases} \qquad (s\bar{t})\langle \bar{u}/x\rangle \coloneqq \sum_{\bar{u}=\bar{v}\cdot\bar{w}} (s\langle \bar{v}/x\rangle)(\bar{t}\langle \bar{w}/x\rangle),$$

$$y\langle \bar{u}/x\rangle \coloneqq \begin{cases} y, & \text{if } \bar{u}=1_Y,\\ \mathbf{0}_{\{y\}}, & \text{otherwise,}\end{cases} \qquad 1_X\langle \bar{u}/x\rangle \coloneqq \begin{cases} 1_{X\{Y/x\}}, & \text{if } \bar{u}=1_Y,\\ \mathbf{0}_{X\{Y/x\}}, & \text{otherwise,}\end{cases}$$

$$(\lambda y.s)\langle \bar{u}/x\rangle \coloneqq \lambda y.s\langle \bar{u}/x\rangle, \qquad ([s]\cdot\bar{t})\langle \bar{u}/x\rangle \coloneqq \sum_{\bar{u}=\bar{v}\cdot\bar{w}} [s\langle \bar{v}/x\rangle]\cdot(\bar{t}\langle \bar{w}/x\rangle).$$

with $x \neq y$ and in the abstraction case $y \notin Y$. We extend it to sums in $\mathbb{N}[(!)\Delta_{\mathsf{I}}(X)]$ by setting:

$$\mathbf{0}_X\langle \bar{u}/x\rangle \coloneqq \mathbf{0}_{X\{Y/x\}} \qquad (s+\mathbf{s})\langle \bar{u}/x\rangle \coloneqq s\langle \bar{u}/x\rangle + \mathbf{s}\langle \bar{u}/x\rangle$$
$$s\langle \mathbf{0}_Y/x\rangle \coloneqq \mathbf{0}_{X\{Y/x\}} \qquad s\langle (\bar{u}+\bar{\mathbf{u}})/x\rangle \coloneqq s\langle \bar{u}/x\rangle + s\langle \bar{\mathbf{u}}/x\rangle.$$

It is easy to verify that the definition above does indeed define a resource sum in $\mathbb{N}[(!)\Delta_{\mathsf{I}}(X\{Y/x\})]$, and that it is stable under the quotients of Definition 24.

Definition 27. The linear degree $\deg_x(s)$ of $x \in \mathcal{V}$ in some $s \in (!)\Delta_{\mathsf{I}}$ is defined by:

$$\deg_x(x) \coloneqq 1, \quad \deg_x(\lambda y.s) \coloneqq \deg_x(s), \text{ wlog. } x \neq y, \quad \deg_x(1_X) \coloneqq 0,$$
$$\deg_x(y) \coloneqq 0, \quad \deg_x(s\bar{t}) \coloneqq \deg_x(s)+\deg_x(\bar{t}), \quad \deg_x([s]\cdot\bar{t}) \coloneqq \deg_x(s)+\deg_x(\bar{t}).$$

The next lemma is not strictly needed, but helps understanding resource substitution.

Lemma 28. Consider $s \in (!)\Delta_{\mathsf{I}}(X)$, $x \in \mathcal{V}$ and $\bar{u} \in !\Delta_{\mathsf{I}}(Y)$, and write $n \coloneqq \deg_x(s)$. Then

$$s\langle \bar{u}/x\rangle = \begin{cases} \sum_{\sigma\in\mathfrak{S}(n)} s\left[u_{\sigma(1)}/x^{(1)},\dots,u_{\sigma(n)}/x^{(n)}\right]\{Y/x\}, & \text{if the cardinality of } \bar{u} \text{ is } n,\\ \mathbf{0}_{X\{Y/x\}}, & \text{otherwise,}\end{cases}$$

where: $\mathfrak{S}(n)$ is the set of permutations of $\{1,\dots,n\}$; $u_1,\dots,u_n$ is any enumeration of the elements in $\bar{u}$; $x^{(1)},\dots,x^{(n)}$ enumerate the occurrences of x in s; and $s\left[u_{\sigma(1)}/x^{(1)},\dots,u_{\sigma(n)}/x^{(n)}\right]$ is the λI-resource term obtained by substituting each element $u_{\sigma(i)}$ for the occurrence $x^{(i)}$.

Proof. Straightforward induction on s. □

The next Substitution Lemma concerns the commutation of resource substitutions, and is the analogous of [4, Lemma 2.1.16]. Notice that the assumptions on the substituted variables are stronger than in the usual resource calculus [21, Lemma 2], where only $x \notin Y$ is required.

Lemma 29. Given $s \in (!)\Delta_{\mathsf{I}}(X)$, $\bar{u} \in !\Delta_{\mathsf{I}}(Y)$, $\bar{v} \in !\Delta_{\mathsf{I}}(Z)$, and $x \in X-Z$, $y \in X \cup Y$,

$$s\langle \bar{u}/x\rangle\langle \bar{v}/y\rangle = \sum_{\bar{v}=\bar{v}'\cdot\bar{v}''} s\langle \bar{v}'/y\rangle\langle \bar{u}\langle \bar{v}''/y\rangle/x\rangle.$$

Proof. By structural induction on s, using Lemma 28 in order to apply the induction hypothesis only to those subterms where the hypotheses are actually satisfied. □

3.3. **The operational semantics.** We finally endow resource expressions and resource sums with a notion of reduction $\to_{\mathsf{r}}$.

Definition 30.

(i) For each $X \subseteq_{\mathrm{f}} \mathcal{V}$, define the resource reduction as a relation between $\lambda\mathsf{I}$-resource terms and resource sums, i.e. $\to_{\mathsf{r}} \subseteq (!)\Delta_{\mathsf{I}}(X) \times \mathbb{N}[(!)\Delta_{\mathsf{I}}(X)]$:

$$\frac{}{(\lambda x.s)\bar{t} \to_{\mathsf{r}} s\,\langle \bar{t}/x\rangle} \qquad \frac{s \to_{\mathsf{r}} \mathbf{s}'}{\lambda x.s \to_{\mathsf{r}} \lambda x.\mathbf{s}'} \qquad \frac{s \to_{\mathsf{r}} \mathbf{s}'}{s\bar{t} \to_{\mathsf{r}} \mathbf{s}'\bar{t}} \qquad \frac{\bar{t} \to_{\mathsf{r}} \bar{\mathbf{t}}'}{s\bar{t} \to_{\mathsf{r}} s\bar{\mathbf{t}}'} \qquad \frac{s \to_{\mathsf{r}} \mathbf{s}'}{[s] \cdot \bar{t} \to_{\mathsf{r}} [\mathbf{s}'] \cdot \bar{t}}$$

(ii) We extend the reduction relation $\to_{\mathsf{r}}$ to resource sums $\mathbb{N}[(!)\Delta_{\mathsf{I}}(X)] \times \mathbb{N}[(!)\Delta_{\mathsf{I}}(X)]$, and simultaneously introduce its reflexive closure $\to_{\mathsf{r}}^{?}$, as follows:

$$\frac{s \to_{\mathsf{r}} \mathbf{s}' \quad \mathbf{t} \to_{\mathsf{r}}^{?} \mathbf{t}'}{s + \mathbf{t} \to_{\mathsf{r}} \mathbf{s}' + \mathbf{t}'} \qquad \frac{\mathbf{t} \to_{\mathsf{r}} \mathbf{t}'}{\mathbf{t} \to_{\mathsf{r}}^{?} \mathbf{t}'} \qquad \frac{}{\mathbf{t} \to_{\mathsf{r}}^{?} \mathbf{t}}$$

(iii) We denote by $\twoheadrightarrow_{\mathsf{r}}$ the reflexive and transitive closure of the relation $\to_{\mathsf{r}}$ given in (ii).

Observe that $\to_{\mathsf{r}}$ is well-defined because $\mathrm{fv}((\lambda x.s)\bar{t}) = \mathrm{fv}(s\,\langle \bar{t}/x\rangle)$. Indeed $\mathrm{fv}((\lambda x.s)\bar{t}) = \mathrm{fv}(s) - \{x\} \cup \mathrm{fv}(\bar{t})$ with $x \in \mathrm{fv}(s)$, whence $\mathrm{fv}(s\,\langle \bar{t}/x\rangle) = \mathrm{fv}(s)\,\{\mathrm{fv}(\bar{t})/x\} = \mathrm{fv}(s) - \{x\} \cup \mathrm{fv}(\bar{t})$.

Example 31. We use the resource $\lambda\mathsf{I}$-terms I and D_n from Example 23.

(i) $\mathsf{D}_0[z] \to_{\mathsf{r}} (x1_{\{x\}})\langle [z]/x\rangle = (x\langle [z]/x\rangle)(1_{\{z\}}) + (x\langle 1_{\{z\}}/x\rangle)(1_{\{x\}}\langle [z]/x\rangle) = z1_{\{z\}} + \mathbf{0}_{\{z\}} = z1_{\{z\}}$. Similarly, $\mathsf{D}_0[\mathsf{I}] \to_{\mathsf{r}} \mathsf{I}1_{\varnothing} \to \mathbf{0}_{\varnothing}$.

(ii) $(\lambda x.\mathsf{D}_0[x])[z]$ has two redexes. Contracting the outermost first gives $(\lambda x.\mathsf{D}_0[x])[z] \to_{\mathsf{r}} \mathsf{D}_0[z] \to_{\mathsf{r}} z1_{\{z\}}$. Contracting the innermost $(\lambda x.\mathsf{D}_0[x])[z] \to_{\mathsf{r}} (\lambda x.x1_{\{x\}})[z] \to_{\mathsf{r}} z1_{\{z\}}$.

(iii) $\mathsf{D}_1[\mathsf{I},\mathsf{I}] \to_{\mathsf{r}} \mathsf{I}[\mathsf{I}] + \mathsf{I}[\mathsf{I}] \to_{\mathsf{r}} \mathsf{I} + \mathsf{I}[\mathsf{I}] \to_{\mathsf{r}} \mathsf{I} + \mathsf{I} = 2.\mathsf{I}$. Thus, sums can arise from single terms.

We show that $\to_{\mathsf{r}}$ enjoys the properties of strong confluence and strong normalisation.

Theorem 32.

(i) The reduction $\to_{\mathsf{r}}$ is strongly normalising.

(ii) $\to_{\mathsf{r}}$ is strongly confluent in the following sense: for all $\mathbf{s}, \mathbf{t}_1, \mathbf{t}_2 \in \mathbb{N}[(!)\Delta_{\mathsf{I}}(X)]$ such that $\mathbf{s} \to_{\mathsf{r}} \mathbf{t}_1$ and $\mathbf{s} \to_{\mathsf{r}} \mathbf{t}_2$, there exists $\mathbf{u} \in \mathbb{N}[(!)\Delta_{\mathsf{I}}(X)]$ such that $\mathbf{t}_1 \to_{\mathsf{r}}^{?} \mathbf{u}$ and $\mathbf{t}_2 \to_{\mathsf{r}}^{?} \mathbf{u}$.

Proof.

(i) The size $\mathsf{size}(s) \in \mathbb{N}$ of a resource expression $s \in (!)\Delta_{\mathsf{I}}$ is defined by structural induction as usual, with base cases $\mathsf{size}(x) = 1$ and $\mathsf{size}(1_X) = 0$, so that $\mathsf{size}(1_X \cdot \bar{t}) = \mathsf{size}(\bar{t})$. The sum-size of a resource sum $\mathbf{s}$ is the finite multiset defined by $\mathsf{ssize}(\mathbf{0}_X) = []$ and $\mathsf{ssize}(s + \mathbf{t}) = [\mathsf{size}(s)] \cdot \mathsf{ssize}(\mathbf{t})$. Sum-sizes are well-ordered by the usual well-founded ordering $\prec_{\mathbb{N}}$ on finite multisets over $\mathbb{N}$ (see [43, §A.6]).

Now, assume that $\mathbf{s} \to_{\mathsf{r}} \mathbf{s}'$. Since the contraction of a redex suppresses an abstraction and there is no duplication, we get $\mathsf{ssize}(\mathbf{s}) \prec_{\mathbb{N}} \mathsf{ssize}(\mathbf{s}')$. Conclude since $\prec_{\mathbb{N}}$ is well-founded.

(ii) The sequence of lemmas leading to the result follows a well-trodden path in the resource calculus [46, 10]. It is sufficient to check that they generalize to our setting. □

By (ii) above every $\mathrm{s} \in \mathbb{N}[(!)\Delta_{\mathsf{I}}(X)]$ has a unique r-normal form which is denoted $\mathrm{nf}(\mathrm{s})$.

4. Taylor approximation for Ohana trees

In its original formulation, Ehrhard and Regnier's Taylor expansion translates a λ-term as a power series of iterated derivatives [20]. We now introduce a qualitative Taylor expansion [34] specifically designed for the $\lambda\mathsf{I}$-calculus, having as target the $\lambda\mathsf{I}$-resource calculus.

4.1. **The Taylor approximation.** Intuitively, a qualitative Taylor expansion should associate each term $M \in \Lambda_{\mathsf{I}}(X)$ with a set of resource approximants $\mathcal{T}_{\mathsf{m}}(M) \subseteq \Delta_{\mathsf{I}}(X)$. Therefore the codomain of $\mathcal{T}_{\mathsf{m}}(-)$ should be $\coprod_{X \subseteq_{\mathrm{f}} \mathcal{V}} 2^{\Delta_{\mathsf{I}}(X)}$, and what we actually define is $\mathcal{T}_{\mathsf{m}}(M) \coloneqq (X, \mathcal{T}^{\bullet}_{\mathsf{m}}(M))$ with $\mathcal{T}^{\bullet}_{\mathsf{m}}(M) \subseteq \Delta_{\mathsf{I}}(X)$. Notice that, whereas all previous constructions of disjoint unions $\coprod_{X \subseteq_{\mathrm{f}} \mathcal{V}}$ were formed from genuinely disjoint sets, this is no longer the case here, as $\varnothing \in 2^{\Delta_{\mathsf{I}}(X)}$, for all X.

Notation 33.

(i) For $X \subseteq_{\mathrm{f}} \mathcal{V}$ and $\mathcal{X} \in 2^{\Delta_{\mathsf{I}}(X)}$, we write $\mathrm{fv}(X, \mathcal{X}) \coloneqq X$.
(ii) We let $\subseteq^{\bullet}$ denote the order relation on $\coprod_{X \subseteq_{\mathrm{f}} \mathcal{V}} 2^{\Delta_{\mathsf{I}}(X)}$ such that $(X, \mathcal{X}) \subseteq^{\bullet} (Y, \mathcal{Y})$ whenever $X = Y$ and $\mathcal{X} \subseteq \mathcal{Y}$. We write $\cup^{\bullet}$ for the corresponding least upper bounds.
(iii) We write $s \in^{\bullet} (X, \mathcal{X})$ to mean that $s \in \mathcal{X}$.
(iv) Given $\mathcal{X} \in 2^{\Delta_{\mathsf{I}}(X)}$, $\mathcal{Y} \in 2^{!\Delta_{\mathsf{I}}(Y)} - \{\varnothing\}$ and $x \in X$, we write:

$$\lambda x.(X, \mathcal{X}) \coloneqq (X - \{x\}, \{\lambda x.s \mid s \in \mathcal{X}\}), \qquad (X, \mathcal{X})\mathcal{Y} \coloneqq (X \cup Y, \{s\bar{t} \mid s \in \mathcal{X},\ \bar{t} \in \mathcal{Y}\}).$$

Definition 34.

(i) The Taylor expansion $\mathcal{T}_{\mathsf{m}}(M) \in \coprod_{X \subseteq_{\mathrm{f}} \mathcal{V}} 2^{\Delta_{\mathsf{I}}(X)}$ of a $\lambda\mathsf{I}$-term M, is defined together with $\mathcal{T}^{!}_{\mathsf{m}}(M) \subset !\Delta_{\mathsf{I}}(\mathrm{fv}(M))$ by mutual induction:

$$\mathcal{T}_{\mathsf{m}}(x) \coloneqq (\{x\}, \{x\}), \quad \mathcal{T}_{\mathsf{m}}(\lambda x.M) \coloneqq \lambda x.\mathcal{T}_{\mathsf{m}}(M), \quad \mathcal{T}_{\mathsf{m}}(MN) \coloneqq \mathcal{T}_{\mathsf{m}}(M)\mathcal{T}^{!}_{\mathsf{m}}(N),$$
$$\mathcal{T}^{!}_{\mathsf{m}}(M) \coloneqq \{1_{\mathrm{fv}(M)}\} \cup \{[t_0, \dots, t_n] \mid n \in \mathbb{N},\ t_0, \dots, t_n \in^{\bullet} \mathcal{T}_{\mathsf{m}}(M)\}.$$

(ii) The above definition is extended to Ohana trees by setting, for all $M \in \Lambda_{\mathsf{I}}$,

$$\mathcal{T}_{\mathsf{m}}(\mathrm{OT}(M)) \coloneqq \bigcup_{A \in \mathrm{App}_{\mathsf{m}}(M)}^{\bullet} \mathcal{T}_{\mathsf{m}}(A), \qquad \text{together with } \begin{cases} \mathcal{T}_{\mathsf{m}}(\bot_X) \coloneqq (X, \varnothing), \\ \mathcal{T}^{!}_{\mathsf{m}}(\bot_X) \coloneqq \{1_X\}. \end{cases}$$

Remark 35. For all M, we have $\mathrm{fv}(\mathcal{T}_{\mathsf{m}}(M)) = \mathrm{fv}(M)$. Similarly, $\mathrm{fv}(\mathcal{T}_{\mathsf{m}}(\mathrm{OT}(M))) = \mathrm{fv}(M)$, due to Theorem 16 and the way we ordered $\coprod_{X \subseteq_{\mathrm{f}} \mathcal{V}} 2^{\Delta_{\mathsf{I}}(X)}$.

Example 36.

(i) $\mathcal{T}_{\mathsf{m}}(\mathsf{I}) = (\varnothing, \{\mathsf{I}\})$, $\mathcal{T}_{\mathsf{m}}(\mathsf{D}) = (\varnothing, \{\mathsf{D}_n \mid n \in \mathbb{N}\})$.
(ii) $\mathcal{T}_{\mathsf{m}}(\mathrm{OT}(\mathsf{Y}f)) = (\{f\}, \mathcal{X}_{\mathsf{Y}f})$ and $\mathcal{T}_{\mathsf{m}}(\mathrm{OT}(\blacksquare_l f)) = (\{f, l\}, \mathcal{X}_{\blacksquare_l f})$, where the sets of approximants can be described as the smallest (in fact, unique) subsets of Δ_{I} such that:

$$\mathcal{X}_{\mathsf{Y}f} = \{f1_{\{f\}}\} \cup \{f[t_0, \dots, t_n] \mid n \in \mathbb{N},\ t_0, \dots, t_n \in \mathcal{X}_{\mathsf{Y}f}\},$$
$$\mathcal{X}_{\blacksquare_l f} = \{f1_{\{f,l\}}\} \cup \{f[t_0, \dots, t_n] \mid n \in \mathbb{N},\ t_0, \dots, t_n \in \mathcal{X}_{\blacksquare_l f}\}.$$

We now describe how resource reduction acts on Taylor expansions, i.e. on potentially infinite sets of resource expressions.

Notation 37.

(i) Given $\mathbf{s} \in \mathbb{N}[(!)\Delta_{\mathsf{l}}(X)]$, we denote by $|\mathbf{s}| \in \coprod_{X \subseteq_{\mathrm{f}} \mathcal{V}} 2^{\Delta_{\mathsf{l}}(X)}$ its support, defined by $|\mathbf{0}_X| \coloneqq (X, \varnothing)$ and $|s + \mathbf{t}| \coloneqq (X, \{s\}) \cup^{\bullet} |\mathbf{t}|$. Notice that $s \in^{\bullet} |\mathbf{t}|$ whenever s bears a non-zero coefficient in $\mathbf{t}$ (i.e. $\mathbf{t} = s + \mathbf{t}'$, for some $\mathbf{t}'$).

(ii) Given $\mathcal{X} \in 2^{(!)\Delta_{\mathsf{l}}(X)}$, we write $\mathrm{nf}(X, \mathcal{X}) \coloneqq \bigcup^{\bullet}_{s \in \mathcal{X}} |\mathrm{nf}(s)|$. Notice that $\mathrm{fv}(\mathrm{nf}(X, \mathcal{X})) = X$, and that $\mathrm{nf}(X, \mathcal{X}) = (X, \varnothing)$ if and only if $s \twoheadrightarrow_{\mathsf{r}} \mathbf{0}_X$, for all $s \in \mathcal{X}$.

This allows us to state the following theorem, adapting [20, Theorem 2].

Theorem 38 (Commutation)**.** For all $M \in \Lambda_{\mathsf{l}}$, $\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M)) = \mathcal{T}_{\mathsf{m}}(\mathrm{OT}(M))$.

We outline a proof in the following sequence of lemmas, similar to the one in [2]; alternatively, the variants of [20, 21], [38] and [10, 12] could also be adapted.

Lemma 39. For all resource sums $\mathbf{s}$ such that $|\mathbf{s}| \subseteq^{\bullet} \mathcal{T}_{\mathsf{m}}(M)$, there exists a reduction $M \twoheadrightarrow_{\beta} N$ such that $|\mathrm{nf}(\mathbf{s})| \subseteq^{\bullet} \mathcal{T}_{\mathsf{m}}(N)$.

Proof. By induction on the length of the longest reduction $\mathbf{s} \twoheadrightarrow_{\mathsf{r}} \mathrm{nf}(\mathbf{s})$. If it is 0, take $N \coloneqq M$. Otherwise, the longest reduction is $\mathbf{s} = t + \mathbf{u} \to_{\mathsf{r}} \mathbf{t}' + \mathbf{u} \twoheadrightarrow_{\mathsf{r}} \mathrm{nf}(\mathbf{s})$. By firing all redexes in M and $\mathbf{u}$ at the same position as the redex fired in $t \to_{\mathsf{r}} \mathbf{t}'$ (formally we do this by induction on this reduction), we obtain $M \to_{\beta} M'$ and $\mathbf{u} \twoheadrightarrow_{\mathsf{r}} \mathbf{u}'$ such that $|\mathbf{t}' + \mathbf{u}'| \subseteq^{\bullet} \mathcal{T}_{\mathsf{m}}(M')$. By confluence, $\mathbf{t}' + \mathbf{u}' \twoheadrightarrow_{\mathsf{r}} \mathrm{nf}(\mathbf{s})$. We conclude by induction on this (shorter) reduction. □

Lemma 40. If $t \in^{\bullet} \mathcal{T}_{\mathsf{m}}(N)$ is in r-nf, then there exists $A \in \mathrm{App}_{\mathsf{m}}(N)$ such that $t \in^{\bullet} \mathcal{T}_{\mathsf{m}}(A)$.

Proof. By induction on t. An r-nf is also a hnf, hence $t = \lambda \vec{x}.y \bar{t}_1 \cdots \bar{t}_k$ and $N = \lambda \vec{x}.y N_1 \cdots N_k$. For $1 \leq i \leq k$, $\bar{t}_i \in \mathcal{T}^!_{\mathsf{m}}(N_i)$. If $\bar{t}_i = 1_{\mathrm{fv}(N_i)}$ define $A_i \coloneqq \bot_{\mathrm{fv}(N_i)}$. If $\bar{t}_i = [t_i^1, \dots, t_i^n]$, each t_i^j is in r-nf, by induction there is $A_i^j \in \mathrm{App}_{\mathsf{m}}(N_i)$, $t_i^j \in^{\bullet} \mathcal{T}_{\mathsf{m}}(A_i^j)$. Define $A_i \coloneqq \bigsqcup_{j=1}^n A_i^j \in \mathrm{App}_{\mathsf{m}}(N_i)$. Finally, $A \coloneqq \lambda \vec{x}.y A_1 \cdots A_k \in \mathrm{App}_{\mathsf{m}}(N)$ and $t \in^{\bullet} \mathcal{T}_{\mathsf{m}}(A)$. □

Lemma 41 (Monotonicity of $\mathcal{T}_{\mathsf{m}}(-)$)**.**

(i) $\mathcal{T}_{\mathsf{m}}(A) \subseteq^{\bullet} \mathcal{T}_{\mathsf{m}}(A')$ if and only if $A \sqsubseteq A'$.

(ii) For all $N \in \Lambda_{\mathsf{l}}$, we have $\mathcal{T}_{\mathsf{m}}(\omega_{\mathsf{m}}(N)) \subseteq^{\bullet} \mathcal{T}_{\mathsf{m}}(N)$.

Proof. Immediate inductions (i) on A and A', and (ii) on the head structure of N. □

Lemma 42 (Simulation of $\to_{\beta}$)**.** If $M \to_{\beta} N$, then $\mathcal{T}_{\mathsf{m}}(N) = \bigcup^{\bullet}_{s \in^{\bullet} \mathcal{T}_{\mathsf{m}}(M)} |\mathbf{t}_s|$ for resource sums $\mathbf{t}_s \in \mathbb{N}[\Delta_{\mathsf{l}}(\mathrm{fv}(M))]$ such that $\forall s \in^{\bullet} \mathcal{T}_{\mathsf{m}}(M),\ s \twoheadrightarrow_{\mathsf{r}} \mathbf{t}_s$.

Proof. One first shows by induction on P that for all $P, Q \in \Lambda_{\mathsf{l}}$ and $x \in \mathrm{fv}(P)$, $\mathcal{T}_{\mathsf{m}}(P[Q/x]) = \bigcup^{\bullet}_{s \in^{\bullet} \mathcal{T}_{\mathsf{m}}(P)} \bigcup^{\bullet}_{\bar{t} \in \mathcal{T}^!_{\mathsf{m}}(Q)} |s \langle \bar{t}/x \rangle|$. Then the proof is an induction on $M \to_{\beta} N$, using the previous equality in the base case. See the details of a similar proof in [12, Lemmas 4.1 and 4.2]. □

Lemma 43. If $A \in \mathcal{A}_{\mathsf{m}}$ then all $s \in^{\bullet} \mathcal{T}_{\mathsf{m}}(A)$ are in r-nf.

Proof. Immediate induction on A. □

Everything is now in place to prove the Commutation Theorem 38.

Proof of Theorem 38. By Theorem 35, $\mathrm{fv}(\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M))) = \mathrm{fv}(M) = \mathrm{fv}(\mathcal{T}_{\mathsf{m}}(\mathrm{OT}(M)))$. Thus, it is sufficient to prove that, for $t \in \Delta_{\mathsf{I}}(\mathrm{fv}(M))$: $t \in^{\bullet} \mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M))$ if and only if $t \in^{\bullet} \mathcal{T}_{\mathsf{m}}(\mathrm{OT}(M))$.

Take $t \in^{\bullet} \mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M))$, i.e. $t \in |\mathrm{nf}(s)|$ for some $s \in^{\bullet} \mathcal{T}_{\mathsf{m}}(M)$. Then, by Theorem 39, there exists a reduction $M \twoheadrightarrow_{\beta} N$ such that $|\mathrm{nf}(s)| \subseteq^{\bullet} \mathcal{T}_{\mathsf{m}}(N)$. Hence $t \in^{\bullet} \mathcal{T}_{\mathsf{m}}(N)$ and t is in r-nf: Theorem 40 ensures that $t \in^{\bullet} \mathcal{T}_{\mathsf{m}}(A)$ for some $A \in \mathrm{App}_{\mathsf{m}}(N) = \mathrm{App}_{\mathsf{m}}(M)$ (by Theorem 19). This means exactly that $t \in^{\bullet} \mathcal{T}_{\mathsf{m}}(\mathrm{OT}(M))$.

Conversely, take $t \in^{\bullet} \mathcal{T}_{\mathsf{m}}(\mathrm{OT}(M))$, i.e. $t \in^{\bullet} \mathcal{T}_{\mathsf{m}}(A)$ for some $A \in \mathrm{App}_{\mathsf{m}}(M)$. By definition there is a reduction $M \twoheadrightarrow_{\beta} N$ such that $A \sqsubseteq \omega_{\mathsf{m}}(N)$. By Theorem 41, $t \in^{\bullet} \mathcal{T}_{\mathsf{m}}(A) \subseteq^{\bullet} \mathcal{T}_{\mathsf{m}}(\omega_{\mathsf{m}}(N)) \subseteq^{\bullet} \mathcal{T}_{\mathsf{m}}(N)$. By iterated applications of Theorem 42, there is an $s \in^{\bullet} \mathcal{T}_{\mathsf{m}}(M)$ such that $s \twoheadrightarrow_{\mathsf{r}} \mathbf{t}_s$ and $t \in |\mathbf{t}_s|$. By Theorem 43, t is in r-nf, therefore $t \in^{\bullet} |\mathrm{nf}(s)| \subseteq^{\bullet} \mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M))$. □

4.2. **The $\lambda\mathsf{I}$-theory of Ohana trees.** Consider the equivalence $\mathcal{O}$ on Λ_{I}, defined by $M =_{\mathcal{O}} N$ if and only if $\mathrm{OT}(M) = \mathrm{OT}(N)$. Thanks to Theorem 38, we are now able to show that this equivalence is a $\lambda\mathsf{I}$-theory.

Corollary 44. For $M, N \in \Lambda_{\mathsf{I}}$, $M =_{\mathcal{O}} N$ if and only if $\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M)) = \mathrm{nf}(\mathcal{T}_{\mathsf{m}}(N))$.

Proof. ($\Rightarrow$) Immediate by Theorem 38. ($\Leftarrow$) Take M, N such that $\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M)) = \mathrm{nf}(\mathcal{T}_{\mathsf{m}}(N))$. By Theorem 20, it is sufficient to prove $\mathrm{App}_{\mathsf{m}}(M) = \mathrm{App}_{\mathsf{m}}(N)$. Take $A \in \mathrm{App}_{\mathsf{m}}(M)$. If $A = \bot_X$, then $A \in \mathrm{App}_{\mathsf{m}}(N)$ is trivial since $\mathrm{App}_{\mathsf{m}}(N)$ is downward closed. Otherwise, by hypothesis and Theorem 38, $\mathcal{T}_{\mathsf{m}}(A) \subseteq^{\bullet} \bigcup^{\bullet}_{B \in \mathrm{App}_{\mathsf{m}}(N)} \mathcal{T}_{\mathsf{m}}(B)$. There is a $B \in \mathrm{App}_{\mathsf{m}}(N)$ such that $[\![A]\!]_{\mathsf{r}} \in^{\bullet} \mathcal{T}_{\mathsf{m}}(B)$, where $[\![A]\!]_{\mathsf{r}} \in^{\bullet} \mathcal{T}_{\mathsf{m}}(A)$ is defined by

$$\begin{array}{rcl} [\![x]\!]_{\mathsf{r}} & \coloneqq & x, \\ [\![\lambda\vec{x}.yA_1 \cdots A_k]\!]_{\mathsf{r}} & \coloneqq & \lambda\vec{x}.y[\![A_1]\!]^{!}_{\mathsf{r}} \cdots [\![A_k]\!]^{!}_{\mathsf{r}}, \end{array} \qquad [\![A]\!]^{!}_{\mathsf{r}} \coloneqq \begin{cases} 1_X, & \text{if } A = \bot_X, \\ [[\![A]\!]_{\mathsf{r}}], & \text{otherwise.} \end{cases}$$

By induction on A, one shows that $[\![A]\!]_{\mathsf{r}} \in^{\bullet} \mathcal{T}_{\mathsf{m}}(B)$ implies $A \sqsubseteq B$, hence $A \in \mathrm{App}_{\mathsf{m}}(N)$ by downward-closure (Theorem 18(ii)). This shows that $\mathrm{App}_{\mathsf{m}}(M) \subseteq \mathrm{App}_{\mathsf{m}}(N)$. The converse inclusion is symmetric. □

Corollary 45. $\mathcal{O}$ is a $\lambda\mathsf{I}$-theory.

Proof. The relation $=_{\mathcal{O}}$ is clearly an equivalence. By Theorems 19 and 20, for all $M, N \in \Lambda_{\mathsf{I}}$, $M =_{\beta} N$ entails $M =_{\mathcal{O}} N$.

For compatibility with abstraction, take $M =_{\mathcal{O}} N$ and $x \in \mathrm{fv}(M) \cap \mathrm{fv}(N)$. By Theorem 44, we get $\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(\lambda x.M)) = \lambda x.\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M)) = \lambda x.\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(N)) = \mathrm{nf}(\mathcal{T}_{\mathsf{m}}(\lambda x.N))$.

For compatibility with application, by Theorem 32 observe that for all $M, N \in \Lambda_{\mathsf{I}}$, $\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(MN)) = \mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M)\mathcal{T}^{!}_{\mathsf{m}}(N)) = \mathrm{nf}(\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M))\mathrm{nf}(\mathcal{T}^{!}_{\mathsf{m}}(N)))$, where $\mathrm{nf}(\mathcal{T}^{!}_{\mathsf{m}}(N))$ is defined by adapting Theorem 37 to sets of resource bags. Therefore, $\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M)) = \mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M'))$ and $\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(N)) = \mathrm{nf}(\mathcal{T}_{\mathsf{m}}(N'))$ imply $\mathrm{nf}(\mathcal{T}_{\mathsf{m}}(MN)) = \mathrm{nf}(\mathcal{T}_{\mathsf{m}}(M'N'))$, and we can conclude the proof by Theorem 44. □

5. Proofs of Section 2

Lemma 14 (recalled from Page 5)**.** There is a bijection between $\mathcal{A}_{\mathsf{m}}$ and the set of finite Ohana trees of $\lambda\mathsf{I}$-terms.

Proof. Given $A \in \mathcal{A}_{\mathsf{m}}$, define a $\lambda\mathsf{I}$-term M_A by structural induction on A:

- if $A = \bot_{\{x_1,\ldots,x_n\}}$ then $M_A \coloneqq \Omega x_1 \cdots x_n$.

- if $A = \lambda\vec{x}.yA_1 \cdots A_k$ then $M_A \coloneqq \lambda\vec{x}.y(M_{A_1}) \cdots (M_{A_k})$.

Define a map ι associating to A the Ohana tree $\mathrm{OT}(M_A)$, i.e. $\iota(A) \coloneqq \mathrm{OT}(M_A)$.

Conversely, let $M \in \Lambda_{\mathsf{I}}$ be such that $T \coloneqq \mathrm{OT}(M)$ is finite. Since T is finite, we can construct an approximant $\iota^-(T)$ by structural induction on T. There are two cases:

- $T = \bot_X$, with $X = \mathrm{fv}(M)$. In this case, define $\iota^-(T) \coloneqq \bot_X$.
- If $T = \lambda\vec{x}.y\cdot^{X_1} T_1 \cdots^{X_k} T_k$ then $M \twoheadrightarrow_h \lambda\vec{x}.yM_1 \cdots M_k$ with $T_i = \mathrm{OT}(M_i)$ and $X = \mathrm{fv}(M_i)$, for all $i\,(1 \leq i \leq k)$. In this case, define $\iota^-(T) = \lambda\vec{x}.y(\iota^-(T_1)) \cdots (\iota^-(T_k))$.

A straightforward induction shows $\iota(\iota^-(T)) = T$ and $\iota^-(\iota(A)) = A$. □

Lemma 18 (recalled from Page 6)**.**

(i) For every $M, N \in \Lambda_{\mathsf{I}}$, $M \to_\beta N$ entails $\omega_{\mathsf{m}}(M) \sqsubseteq \omega_{\mathsf{m}}(N)$.
(ii) For all $M \in \Lambda_{\mathsf{I}}(X)$, $\mathrm{App}_{\mathsf{m}}(M)$ is an ideal of the poset $(\mathcal{A}_{\mathsf{m}}(X), \sqsubseteq)$. More precisely:
  (a) non-emptiness: $\bot_X \in \mathrm{App}_{\mathsf{m}}(M)$;
  (b) downward closure: $A \in \mathrm{App}_{\mathsf{m}}(M)$ and $A' \sqsubseteq A$ imply $A' \in \mathrm{App}_{\mathsf{m}}(M)$;
  (c) directedness: $\forall A_1, A_2 \in \mathrm{App}_{\mathsf{m}}(M), \exists A_3 \in \mathrm{App}_{\mathsf{m}}(M)$ such that $A_1 \sqsubseteq A_3 \sqsupseteq A_2$.

Proof. (i) By induction on a derivation of $M \to_\beta N$. By Theorem 2, either M has a head redex or it has a head variable. If $M = \lambda\vec{x}.(\lambda y.M')M_0M_1 \cdots M_k$ then $\omega_{\mathsf{m}}(M) = \bot_{\mathrm{fv}(M)}$. By Theorem 7(ii), $M \to_\beta N$ entails $\mathrm{fv}(N) = \mathrm{fv}(M)$, whence $\omega_{\mathsf{m}}(N) \in \mathcal{A}_{\mathsf{m}}(\mathrm{fv}(M))$. This case follows since $\bot_{\mathrm{fv}(M)}$ is the bottom of $\mathcal{A}_{\mathsf{m}}(\mathrm{fv}(M))$.

If $M = \lambda\vec{x}.yM_1 \cdots M_k$ then $\omega_{\mathsf{m}}(M) = \lambda\vec{x}.y\,\omega_{\mathsf{m}}(M_1) \cdots \omega_{\mathsf{m}}(M_k)$. The β-reduction must occur in some $M_i \to_\beta N_i$, whence $\omega_{\mathsf{m}}(N) = \lambda\vec{x}.y\,\omega_{\mathsf{m}}(M_1) \cdots \omega_{\mathsf{m}}(N_i) \cdots \omega_{\mathsf{m}}(M_k)$. We conclude since $\omega_{\mathsf{m}}(M_i) \sqsubseteq \omega_{\mathsf{m}}(N_i)$ holds, from the induction hypothesis.

(ii) To prove that $\mathrm{App}_{\mathsf{m}}(M)$ is an ideal, we need to check three conditions.
  (a) non-emptiness: since $\bot_X \sqsubseteq \omega_{\mathsf{m}}(M)$ for all $M \in \mathcal{A}_{\mathsf{m}}(X)$, we get $\bot_X \in \mathrm{App}_{\mathsf{m}}(M)$.
  (b) downward closure: by definition of $\mathrm{App}_{\mathsf{m}}(M)$.
  (c) directedness: let us take $A_1, A_2 \in \mathrm{App}_{\mathsf{m}}(M)$ and prove that $A_1 \sqcup A_2 \in \mathrm{App}_{\mathsf{m}}(M)$. We proceed by structural induction on, say, A_1. Case $A_1 = \bot_X$. Then $A_2 \in \mathcal{A}_{\mathsf{m}}(X)$ and $A_1 \sqcup A_2 = A_2 \in \mathrm{App}_{\mathsf{m}}(M)$. Case $A_1 = \lambda x_1 \ldots x_n.y\,A'_1 \cdots A'_k$. If $A_2 = \bot_X$ then proceed as before. Otherwise $A_1 \in \mathrm{App}_{\mathsf{m}}(M)$ because there is a reduction $M \twoheadrightarrow_\beta \lambda x_1 \ldots x_n.y\,M'_1 \cdots M'_k =: M_1$ with $A'_i \in \mathrm{App}_{\mathsf{m}}(M'_i)$. Similarly, $A_2 \in \mathrm{App}_{\mathsf{m}}(M)$ because there is a reduction $M \twoheadrightarrow_\beta \lambda x_1 \ldots x_{n'}.y\,M''_1 \cdots M''_{k'} =: M_2$ with $A_2 = \lambda x_1 \ldots x_{n'}.y\,A''_1 \cdots A''_{k'}$ and $A''_j \in \mathrm{App}_{\mathsf{m}}(M''_j)$. By confluence of $\to_\beta$, $n = n'$ and $k = k'$, and every M'_i, M''_i have a common reduct M'''_i. By Theorem 18(i), $A'_i, A''_i \in \mathrm{App}_{\mathsf{m}}(M'''_i)$, so by HI $A'_i \sqcup A''_i \in \mathrm{App}_{\mathsf{m}}(M'''_i)$. Conclude by taking $A_3 \coloneqq \lambda x_1 \ldots x_n.y\,(A'_1 \sqcup A''_1) \cdots (A'_k \sqcup A''_k)$. □

Proposition 19 (recalled from Page 6)**.** Let $M, N \in \Lambda_{\mathsf{I}}(X)$. If $M \to_\beta N$, then $\mathrm{App}_{\mathsf{m}}(M) = \mathrm{App}_{\mathsf{m}}(N)$.

Proof. To prove $\mathrm{App}_{\mathsf{m}}(M) = \mathrm{App}_{\mathsf{m}}(N)$, we show the two inclusions.

($\subseteq$) Take $A \in \mathrm{App}_{\mathsf{m}}(M)$, then there exists a reduction $M \twoheadrightarrow_\beta M'$, such that $A \sqsubseteq \omega_{\mathsf{m}}(M')$. We proceed by induction on A.

Case $A = \bot_X$. By Remark 16, since $\bot_X \sqsubseteq \omega_{\mathsf{m}}(N)$, for all $N \in \Lambda_{\mathsf{I}}(X)$.

Case $A = \lambda\vec{x}.yA_1 \cdots A_k$. Then $M' = \lambda\vec{x}.yM'_1 \cdots M'_k$ with $A_i \sqsubseteq \omega_{\mathsf{m}}(M'_i)$, for all $1 \leq i \leq k$. Since $M \to_\beta N$, by confluence, N and M' have a common reduct $N' = \lambda\vec{x}.yN'_1 \cdots N'_k$ with $M'_i \twoheadrightarrow_\beta N'_i$. By Theorem 18(i) we have $\omega_{\mathsf{m}}(M'_i) \sqsubseteq \omega_{\mathsf{m}}(N'_i)$ and, by transitivity, $A_i \sqsubseteq \omega_{\mathsf{m}}(N'_i)$. We found a reduction $N \twoheadrightarrow_\beta N'$ such that $A \sqsubseteq \omega_{\mathsf{m}}(N')$, whence $A \in \mathrm{App}_{\mathsf{m}}(N)$.

($\supseteq$) Straightforward, since every reduct of N is also a reduct of M. □

For the proof of Theorem 20, we introduce some definitions. First, given an Ohana tree $T = \mathrm{OT}(M)$, one defines $\mathrm{fv}(T)$ to be $\mathrm{fv}(M)$. If $T = \lambda x_1 \ldots x_n.y \cdot^{X_1} T_1 \ldots^{X_k} T_k$. then $\mathrm{fv}(T) = \bigcup_{i=1}^{k} X_i \cup \{y\} - \{x_1, \ldots, x_n\}\}\}$. Moreover, in the following, we will consider the class of head contexts, defined inductively by the following grammar: $\mathsf{H} ::= [] \mid \lambda x_1 \ldots x_n.y\mathsf{H}_1 \cdots \mathsf{H}_k$.

Then, in order to relate Ohana trees with approximants, we extend the order on approximants to Ohana trees and define the set of finite subtrees of an Ohana tree:

Definition 46. Let $\mathcal{M} = \{\mathrm{OT}(M) \mid M \in \Lambda_{\mathsf{I}}\}$. We define $\sqsubseteq \subseteq \mathcal{M}^2$ as the least partial order on Ohana trees closed under the following rules:

$$\frac{\mathrm{fv}(T) = X}{\bot_X \sqsubseteq T} \qquad \frac{T_i \sqsubseteq T'_i \quad \text{for all } 1 \leq i \leq k}{\lambda x_1 \ldots x_n.y \cdot^{X_1} T_1 \ldots^{X_k} T_k \sqsubseteq \lambda x_1 \ldots x_n.y \cdot^{X_1} T'_1 \ldots^{X_k} T'_k}$$

Definition 47. Given an Ohana tree $T = \mathrm{OT}(M)$, one defines $T^\star$ to be $\{T' \mid T' \sqsubseteq T \text{ and } T' \text{ is finite}\}$.

Definition 48. We define inductively the set of positions of an approximant A, $\mathsf{Pos}(A)$ as a subset of $\mathbb{N}^\star$ such that:

- If $A = \bot_X$, $\mathsf{Pos}(A) = \epsilon$;
- If $A = \lambda x_1 \ldots x_n.yA_1 \cdots A_k$, $\mathsf{Pos}(A) = \{i \cdot p \mid 1 \leq i \leq k \,\&\, p \in \mathsf{Pos}(A_i)\}$.

Sets of positions are lifted to sets of approximants as expected and one can similarly define $\mathsf{Pos}(M) = \mathsf{Pos}(\mathrm{App}_{\mathsf{m}}(M))$.

Definition 49. Given an approximant A and $p \in \mathsf{Pos}(A)$, one defines $A(p)$ by induction on the length of p as follows:

- If $p = \epsilon$, then if $A = \bot_X$, $A(p) = \bot_X$, otherwise, $A = \lambda x_1 \ldots x_n.yA_1 \cdots A_k$ and $A(p) = \lambda x_1 \ldots x_n.y$;
- If $p = i \cdot p'$, then $A = \lambda x_1 \ldots x_n.yA_1 \cdots A_k$, with $1 \leq i \leq k$ and $A(p) = A_i(p')$.

Theorem 20 (recalled from Page 6)**.** There is a bijection $\mathcal{A}(\cdot)$ between the set of Ohana trees and the set of approximants of $\lambda\mathsf{I}$-terms satisfying $\mathcal{A}(\mathrm{OT}(M)) = \mathrm{App}_{\mathsf{m}}(M)$, for all $M \in \Lambda_{\mathsf{I}}$.

Proof of Theorem 20. Let us write $\mathbb{T} = \{\mathrm{OT}(M) \mid M \in \Lambda_{\mathsf{I}}\}$ and $\mathbb{A} = \{\mathrm{App}_{\mathsf{m}}(M) \mid M \in \Lambda_{\mathsf{I}}\}$ and let us build a bijection $\mathcal{A} : \mathbb{T} \to \mathbb{A}$ such that for any $M \in \Lambda_{\mathsf{I}}$, $\mathcal{A}(\mathrm{OT}(M)) = \mathrm{App}_{\mathsf{m}}(M)$. Let us define $\mathcal{A}$. Let $M \in \Lambda_{\mathsf{I}}$ and T its Ohana tree. Since $T^\star$ contains only finite "subtrees" of T, which are all Ohana trees of $\lambda\mathsf{I}$-terms, one can consider a set of approximants $\mathcal{A}(T)$ obtained as the direct image of $T^\star$ by the mapping ι^- defined in the proof of Lemma 14.

In other words, we define $\mathcal{A}(T) \coloneqq \iota^-(T^\star)$. We now prove that $\mathcal{A}(T) = \mathrm{App}_{\mathsf{m}}(M)$.

Let us prove that $\mathcal{A}(T) \subseteq \mathrm{App}_{\mathsf{m}}(M)$. Let $T' \in T^\star$, then there exist $M_1 \in \Lambda_{\mathsf{I}}(X_1), \ldots, M_n \in \Lambda_{\mathsf{I}}(X_n)$ and a head context H such that $\mathsf{H}[M_1, \ldots, M_n] \in \Lambda_{\mathsf{I}}$ with $M \twoheadrightarrow_\beta \mathsf{H}[M_1, \ldots, M_n]$ and $T = \mathrm{OT}(\mathsf{H}[\Omega\vec{X_1}, \ldots, \Omega\vec{X_n}])$. Therefore

$\iota^-(T') = \mathsf{H}[\bot_{X_1}, \ldots, \bot_{X_n}]$ and since $\bot_{X_i} \sqsubseteq \omega_{\mathsf{m}}(M_i)$, $\iota^-(T') \in \mathrm{App}_{\mathsf{m}}(M)$ which proves the inclusion.

For the converse inclusion, we prove by induction on the structure of A that, for any $M \in \Lambda_{\mathsf{l}}$ and $A \in \mathrm{App}_{\mathsf{m}}(M)$, $A \in \mathcal{A}(\mathrm{OT}(M))$:

- If $A = \bot_X$, then $X = \mathrm{fv}(M)$ and $A \in \iota^-(\mathrm{OT}(M)^\star)$;
- Otherwise, $A = \lambda x_1 \ldots x_n.yA_1 \cdots A_k$. In such a case, we have $M \to_h \lambda x_1 \ldots x_n.yM_1 \cdots M_k$ with $M_i \in \Lambda_{\mathsf{l}}(X_i)$ and $A_i \in \mathrm{App}_{\mathsf{m}}(M_i)$ for $1 \leq i \leq k$. By induction hypothesis we have some $T_i \in \mathrm{OT}(M_i)^\star$ such that $A_i \in \iota^-(T_i)$ for all $i\,(1 \leq i \leq k)$. Since $\lambda x_1 \ldots x_n.y \cdot^{X_1} T_1 \cdots^{X_k} T_k \in \mathrm{OT}(M)^\star$, we indeed have that $A \in \mathcal{A}(\mathrm{OT}(M))$.

Let us define $\mathcal{T}$, the inverse of $\mathcal{A}$. Let $A, B \in \mathrm{App}_{\mathsf{m}}(M)$. By directedness of $\mathrm{App}_{\mathsf{m}}(M)$ (Lemma 18(ii)), there exists $C \in \mathrm{App}_{\mathsf{m}}(M)$ such that $A, B \sqsubseteq C$. From this, it follows that at any position where two approximants of M are defined and different from $\bot_X$, they agree. We define a (potentially) infinite tree $\mathcal{T}(\mathrm{App}_{\mathsf{m}}(M))$ from $\mathrm{App}_{\mathsf{m}}(M)$ as a pair of partial functions (v, e), respectively labelling the edges and vertices of the tree, defined on $\mathsf{Pos}(\mathrm{App}_{\mathsf{m}}(M))$ as follows:

- for every position $p \in \mathsf{Pos}(\mathrm{App}_{\mathsf{m}}(M))$, either for every $A \in \mathrm{App}_{\mathsf{m}}(M)$ such that $p \in \mathsf{Pos}(A)$, $A(p) = \bot_X$, in which case $v(p) = \bot_X$, or there is some $A \in \mathrm{App}_{\mathsf{m}}(M)$ such that $p \in \mathsf{Pos}(A)$, $A(p) = \lambda x_1 \ldots x_n.y$ in which case $v(p) \coloneqq \lambda x_1 \ldots x_n.y$.
- for every non-empty position $p \in \mathsf{Pos}(\mathrm{App}_{\mathsf{m}}(M))$, let $A \in \mathrm{App}_{\mathsf{m}}(M)$ be such that $A(p) = \bot_X$ and set $e(p) \coloneqq X$.

A simple coinduction gives us $\mathcal{T}(\mathcal{A}(\mathrm{OT}(M))) = \mathrm{OT}(M)$ for any $M \in \Lambda_{\mathsf{l}}$. □

6. Proofs of Section 3

6.1. **Substitution lemma (Lemma 29).**

Lemma 50. Given $s \in (!)\Delta_{\mathsf{l}}(X)$, $\bar{u} \in\, !\Delta_{\mathsf{l}}(Y)$ and $x \notin X$,

$$s\,\langle \bar{u}/x\rangle = \begin{cases} s, & \text{if } \bar{u} = 1_Y, \\ \mathbf{0}_X, & \text{otherwise.} \end{cases}$$

Proof. Straightforward induction on s. □

Lemma 29 (recalled from Page 9)**.** Given $s \in (!)\Delta_{\mathsf{l}}(X)$, $\bar{u} \in\, !\Delta_{\mathsf{l}}(Y)$, $\bar{v} \in\, !\Delta_{\mathsf{l}}(Z)$, $x \in X - Z$ and $y \in X \cup Y$,

$$s\,\langle \bar{u}/x\rangle\,\langle \bar{v}/y\rangle = \sum_{\bar{v}=\bar{v}'\cdot\bar{v}''} s\,\langle \bar{v}'/y\rangle\,\langle \bar{u}\,\langle \bar{v}''/y\rangle/x\rangle\,.$$

Proof. By induction on s.

- Case $s = x$. (It is the only possible case of a variable because of the hypothesis $x \in X$.) We have

  $$\sum_{\bar{v}=\bar{v}'\cdot\bar{v}''} x\,\langle \bar{v}'/y\rangle\,\langle \bar{u}\,\langle \bar{v}''/y\rangle/x\rangle = x\,\langle 1_Z/y\rangle\,\langle \bar{u}\,\langle \bar{v}/y\rangle/x\rangle = x\,\langle \bar{u}\,\langle \bar{v}/y\rangle/x\rangle\,,$$

  hence if $\bar{u} = [u]$, then both sides of the equality are equal to $u\,\langle \bar{v}/y\rangle$, otherwise they are equal to $\mathbf{0}_{Y\{Z/y\}}$.
- Case $s = \lambda y.s'$. Immediate by the induction hypothesis on s'.
- Case $s = s'\bar{s}''$. We have $X = \mathrm{fv}(s) = \mathrm{fv}(s') \cup \mathrm{fv}(\bar{s}'')$. There are several possible cases.

- If $x \in \mathrm{fv}(s')$ and $x \in \mathrm{fv}(\bar{s}'')$, then it is possible to apply the induction hypothesis in both subterms. By the definition of resource substitution,

$$\begin{aligned} s\,\langle \bar{u}/x\rangle\,\langle \bar{v}/y\rangle &= \sum_{\bar{u}=\bar{u}_l\cdot\bar{u}_r}\ \sum_{\bar{v}=\bar{v}_l\cdot\bar{v}_r} s'\,\langle \bar{u}_l/x\rangle\,\langle \bar{v}_l/y\rangle\,\bar{s}''\,\langle \bar{u}_r/x\rangle\,\langle \bar{v}_r/y\rangle \\ &= \sum_{\bar{u}=\bar{u}_l\cdot\bar{u}_r}\ \sum_{\bar{v}=(\bar{v}'_l\cdot\bar{v}''_l)\cdot(\bar{v}'_r\cdot\bar{v}''_r)} s'\,\langle \bar{v}'_l/y\rangle\,\langle \bar{u}_l\,\langle \bar{v}''_l/y\rangle/x\rangle\,\bar{s}''\,\langle \bar{v}'_r/y\rangle\,\langle \bar{u}_r\,\langle \bar{v}''_r/y\rangle/x\rangle \end{aligned}$$

  by the induction hypotheses on s' and $\bar{s}''$,

$$= \sum_{\bar{v}=\bar{v}'\cdot\bar{v}''} s\,\langle \bar{v}'/y\rangle\,\langle \bar{u}\,\langle \bar{v}''/y\rangle/x\rangle$$

  by permuting the sums and re-arranging the indices (using associativity and commutativity of the multiset union).
- If $x \in \mathrm{fv}(s)$ and $x \notin \mathrm{fv}(\bar{s}'')$, then again

$$\begin{aligned} s\,\langle \bar{u}/x\rangle\,\langle \bar{v}/y\rangle &= \sum_{\bar{u}=\bar{u}_l\cdot\bar{u}_r}\ \sum_{\bar{v}=\bar{v}_l\cdot\bar{v}_r} s'\,\langle \bar{u}_l/x\rangle\,\langle \bar{v}_l/y\rangle\,\bar{s}''\,\langle \bar{u}_r/x\rangle\,\langle \bar{v}_r/y\rangle \\ &= \sum_{\bar{v}=\bar{v}_l\cdot\bar{v}_r} s'\,\langle \bar{u}/x\rangle\,\langle \bar{v}_l/y\rangle\,\bar{s}''\,\langle 1_Y/x\rangle\,\langle \bar{v}_r/y\rangle \\ &\quad + \sum_{\substack{\bar{u}=\bar{u}_l\cdot\bar{u}_r\\ \bar{u}_r\neq 1_Y}}\ \sum_{\bar{v}=\bar{v}_l\cdot\bar{v}_r} s'\,\langle \bar{u}_l/x\rangle\,\langle \bar{v}_l/y\rangle\,\bar{s}''\,\langle \bar{u}_r/x\rangle\,\langle \bar{v}_r/y\rangle \\ &= \sum_{\bar{v}=\bar{v}_l\cdot\bar{v}_r} s'\,\langle \bar{u}/x\rangle\,\langle \bar{v}_l/y\rangle\,\bar{s}''\,\langle \bar{v}_r/y\rangle + \mathbf{0}_{X\{Y/x\}\{Z/y\}} \end{aligned}$$

  by Theorem 50,

$$= \sum_{\bar{v}=(\bar{v}'_l\cdot\bar{v}''_l)\cdot\bar{v}_r} s'\,\langle \bar{v}'_l/y\rangle\,\langle \bar{u}\,\langle \bar{v}''_l/y\rangle/x\rangle\,\bar{s}''\,\langle \bar{v}_r/y\rangle$$

  by the induction hypothesis on s',

$$= \sum_{\bar{v}=(\bar{v}'_l\cdot\bar{v}_r)\cdot\bar{v}''_l} s\,\langle \bar{v}'_l\cdot\bar{v}_r/y\rangle\,\langle \bar{u}\,\langle \bar{v}''_l/y\rangle/x\rangle$$

  by re-arranging the indices, and using again Theorem 50 together with the hypotheses $x \notin Z$ and $x \notin \mathrm{fv}(\bar{s}'')$. This is exactly the desired result.
- The case $x \notin \mathrm{fv}(s)$ and $x \in \mathrm{fv}(\bar{s}'')$ is symmetric.

- Case $s = 1_X$. If $\bar{u} = 1_Y$ and $\bar{v} = 1_Z$, both sides of the equality are equal to $1_{X\{Y/x\}\{Z/y\}}$. Otherwise, they are equal to $\mathbf{0}_{X\{Y/x\}\{Z/y\}}$.
- Case $s = s'\cdot\bar{s}''$ is similar to the case of application above. □

6.2. **Strong confluence of $\to_{\mathsf{r}}$ (Theorem 32, item ii).**

Lemma 51. If $s \to_{\mathsf{r}} \mathbf{s}'$ then $s\,\langle \bar{t}/x\rangle \to_{\mathsf{r}}^{?} \mathbf{s}'\,\langle \bar{t}/x\rangle$.

Proof. By induction on $s \to_{\mathsf{r}} \mathbf{s}'$. For the base case $(\lambda y.u)\bar{v} \to_{\mathsf{r}} u\,\langle \bar{v}/y\rangle$,

$$\begin{aligned} ((\lambda y.u)\bar{v})\,\langle \bar{t}/x\rangle &= \sum_{\bar{t}=\bar{t}'\cdot\bar{t}''} (\lambda y.u\,\langle \bar{t}'/x\rangle)\,\bar{v}\,\langle \bar{t}''/y\rangle \\ &\to_{\mathsf{r}}^{?} \sum_{\bar{t}=\bar{t}'\cdot\bar{t}''} u\,\langle \bar{t}'/x\rangle\,\langle \bar{v}\,\langle \bar{t}''/x\rangle/y\rangle \\ &= u\,\langle \bar{v}/y\rangle\,\langle \bar{t}/x\rangle \qquad \text{by Theorem 29.} \end{aligned}$$

Notice that the reflexive closure $\to_{\mathsf{r}}^{?}$ of $\to_{\mathsf{r}}$ is necessary to handle the case where the two sums are empty. The other cases are immediate applications of the induction hypotheses. □

Lemma 52. If $\bar{t} \to_{\mathsf{r}} \bar{\mathbf{t}}'$ then $s\,\langle \bar{t}/x\rangle \to_{\mathsf{r}}^{?} s\,\langle \bar{\mathbf{t}}'/x\rangle$.

Proof. By a straightforward induction on the different cases defining $s\,\langle \bar{t}/x\rangle$. □

Lemma 53. For all $s \in (!)\Delta_{\mathsf{l}}(X)$ and $\mathbf{s}', \mathbf{s}'' \in \mathbb{N}[(!)\Delta_{\mathsf{l}}(X)]$ such that

$$s \to_{\mathsf{r}} \mathbf{s}' \quad \text{and} \quad s \to_{\mathsf{r}} \mathbf{s}'',$$

there exists $\mathbf{t} \in \mathbb{N}[(!)\Delta_{\mathsf{l}}(X)]$ such that

$$\mathbf{s}' \to_{\mathsf{r}}^{?} \mathbf{t} \quad \text{and} \quad \mathbf{s}'' \to_{\mathsf{r}}^{?} \mathbf{t}.$$

Proof. Take s, $\mathbf{s}'$ and $\mathbf{s}''$ as above. We proceed by induction on both reductions $s \to_{\mathsf{r}} \mathbf{s}'$ and $s \to_{\mathsf{r}} \mathbf{s}''$. The base case is when the first reduction comes from the first rule in Theorem 30, i.e. $s = (\lambda x.u)\bar{v}$ and $\mathbf{s}' = u\,\langle \bar{v}/x\rangle$:

- If the second reduction comes from the same rule, then $\mathbf{s}'' = \mathbf{s}'$ and we set $\mathbf{t} \coloneqq \mathbf{s}'$.
- If the second reduction comes from the rule for left application, i.e. $\mathbf{s}'' = (\lambda x.\mathbf{u}'')\bar{v}$ with $u \to_{\mathsf{r}} \mathbf{u}''$, then by Theorem 51 $s' = u\,\langle \bar{v}/x\rangle \to_{\mathsf{r}}^{?} \mathbf{u}''\,\langle \bar{v}/x\rangle$ and by Theorem 30 $s'' \to_{\mathsf{r}}^{?} \mathbf{u}''\,\langle \bar{v}/x\rangle$. (The reflexive closure is needed in the latter case because $\mathbf{u}''$ may be empty.)
- If the second reduction comes from the rule for right application, then the proof is as in the previous case, using Theorem 52 instead of Theorem 51.

In all other cases, the results immediately follows from the induction hypotheses. □

Theorem 32 (Item ii, recalled from Page 10)**.** For all $\mathbf{s}, \mathbf{s}', \mathbf{s}'' \in \mathbb{N}[(!)\Delta_{\mathsf{l}}(X)]$ such that

$$\mathbf{s} \to_{\mathsf{r}} \mathbf{s}' \quad \text{and} \quad \mathbf{s} \to_{\mathsf{r}} \mathbf{s}'',$$

there exists $\mathbf{t} \in \mathbb{N}[(!)\Delta_{\mathsf{l}}(X)]$ such that

$$\mathbf{s}' \to_{\mathsf{r}}^{?} \mathbf{t} \quad \text{and} \quad \mathbf{s}'' \to_{\mathsf{r}}^{?} \mathbf{t}.$$

Proof. We prove the result under the slightly more general hypothesis that $\mathbf{s} \to_{\mathsf{r}}^{?} \mathbf{s}'$ and $\mathbf{s} \to_{\mathsf{r}}^{?} \mathbf{s}''$. We proceed by induction on $\mathsf{ssize}(\mathbf{s})$. If $\mathsf{ssize}(\mathbf{s}) = []$, i.e. $\mathbf{s}$ is empty, then both reductions must be equalities and the conclusion is trivial. Otherwise, suppose $\mathsf{ssize}(\mathbf{s}) \succ_{\mathbb{N}} []$. Again, if any of the two reductions turns out to be an equality, the result is immediate. Otherwise $\mathbf{s} \to_{\mathsf{r}} \mathbf{s}'$ and $\mathbf{s} \to_{\mathsf{r}} \mathbf{s}''$, i.e.

- $\mathbf{s} = s_1 + \mathbf{s}_{\neq 1}$ and $\mathbf{s}' = \mathbf{s}'_1 + \mathbf{s}'_{\neq 1}$, together with $s_1 \to_{\mathsf{r}} \mathbf{s}'_1$ and $\mathbf{s}_{\neq 1} \to_{\mathsf{r}}^{?} \mathbf{s}'_{\neq 1}$,
- $\mathbf{s} = s_2 + \mathbf{s}_{\neq 2}$ and $\mathbf{s}'' = \mathbf{s}''_2 + \mathbf{s}''_{\neq 2}$, together with $s_2 \to_{\mathsf{r}} \mathbf{s}''_2$ and $\mathbf{s}_{\neq 2} \to_{\mathsf{r}}^{?} \mathbf{s}''_{\neq 2}$.

There are two possible cases.

- If $s_1 = s_2$ and $\mathbf{s}_{\neq 1} = \mathbf{s}_{\neq 2}$, then by Theorem 53 applied to $s_1 \to_{\mathsf{r}} \mathbf{s}'_1$ and $s_1 \to_{\mathsf{r}} \mathbf{s}''_2$, we obtain $\mathbf{t}_1$ such that $\mathbf{s}'_1 \to_{\mathsf{r}} \mathbf{t}_1$ and $\mathbf{s}''_2 \to_{\mathsf{r}} \mathbf{t}_1$. By the induction hypothesis on $\mathbf{s}_{\neq 1}$, we obtain $\mathbf{t}_{\neq 1}$ such that $\mathbf{s}'_{\neq 1} \to_{\mathsf{r}} \mathbf{t}_{\neq 1}$ and $\mathbf{s}''_{\neq 2} \to_{\mathsf{r}} \mathbf{t}_{\neq 1}$. Therefore we can set $\mathbf{t} \coloneqq \mathbf{t}_1 + \mathbf{t}_{\neq 1}$.
- If $s_1 \neq s_2$, then we can write $\mathbf{s} = s_1 + s_2 + \mathbf{s}_{\neq 12}$, with:

$$\begin{array}{llll} \mathbf{s}_{\neq 1} = s_2 + \mathbf{s}_{\neq 12} & \mathbf{s}'_{\neq 1} = \mathbf{s}'_2 + \mathbf{s}'_{\neq 12} & s_2 \to_{\mathsf{r}}^{?} \mathbf{s}'_2 & \mathbf{s}_{\neq 12} \to_{\mathsf{r}}^{?} \mathbf{s}'_{\neq 12} \\ \mathbf{s}_{\neq 2} = s_1 + \mathbf{s}_{\neq 12} & \mathbf{s}''_{\neq 2} = \mathbf{s}''_1 + \mathbf{s}''_{\neq 12} & s_1 \to_{\mathsf{r}}^{?} \mathbf{s}''_1 & \mathbf{s}_{\neq 12} \to_{\mathsf{r}}^{?} \mathbf{s}''_{\neq 12}. \end{array}$$

By Theorem 53 applied to s_1 we obtain $\mathbf{t}_1$ such that $\mathbf{s}'_1 \to_{\mathsf{r}}^{?} \mathbf{t}_1$ and $\mathbf{s}''_1 \to_{\mathsf{r}}^{?} \mathbf{t}_1$. By the same lemma applied to s_2 we obtain $\mathbf{t}_2$ such that $\mathbf{s}'_2 \to_{\mathsf{r}}^{?} \mathbf{t}_2$ and $\mathbf{s}''_2 \to_{\mathsf{r}}^{?} \mathbf{t}_2$. By the induction hypothesis on $\mathbf{s}_{\neq 12}$ we obtain $\mathbf{t}_{\neq 12}$ such that $\mathbf{s}'_{\neq 12} \to_{\mathsf{r}}^{?} \mathbf{t}_{\neq 12}$ and $\mathbf{s}''_{\neq 12} \to_{\mathsf{r}}^{?} \mathbf{t}_{\neq 12}$. Finally, we can set $\mathbf{t} \coloneqq \mathbf{t}_1 + \mathbf{t}_2 + \mathbf{t}_{\neq 12}$. □

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